

Decentralized Crypto World Bank

Heavily Compressed Pre-Thesis (v31)

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Chapter 1

Introduction

The *Crypto World Bank* is a research-based project for exploring how blockchain technology can be shaped to fit into institutional financial systems with all the added benefits that come with a decentralized financing system. Based on our findings that had similar motivation to our study, existing DeFi protocols show isolated functionalities, such as Aave for lending, Uniswap for exchange, and MakerDAO for stablecoins. This project specifies a tier-based, hierarchically governed platform that includes a mixture of institutional financing with DeFi, accommodating deposit mobilization, credit allocation, interbank liquidity, installment repayment, FX facilitation, solidarity group lending, reserve enforcement, syndicated co-lending, and risk-based governance covering all four tiers that systematically mirrors real-world institutional financing. At the top, Tier 1 represents the World Bank reserve; the core banking hierarchy is specified in Solidity and being improved until final deployment. Another significant feature—the multi-entity contract suite—is fully specified in Chapter 3, which is scheduled for practical development in Phases II–III.

The system is designed for three categories of participant: a programmable World Bank reserve as Tier 1, national and local development banks serving in Tiers 2–3, and retail clients in Tier 4. Retail clients get access to savings, group loans, and installment lending features. This hierarchical structure in this project was motivated by unbanked and MSME financing gaps [9–10]. The project is not built with any motivation to deploy only in regions with access to advanced technologies; rather, proper distribution of access to finance and fairness is the leading cause of this attempt to find suitable ground to adapt DeFi among the general population. The whole project is open source and available in our GitHub repository. The added benefits of blockchain: it provides state visibility, tamper-evident audit trails, and programmable rule enforcement among institutions having competition, as demonstrated by the renowned World Bank FundsChain programme and JPMorgan Kinexys. From the study of institutions, open ledger infrastructure and institutional hierarchy are not mutually exclusive.

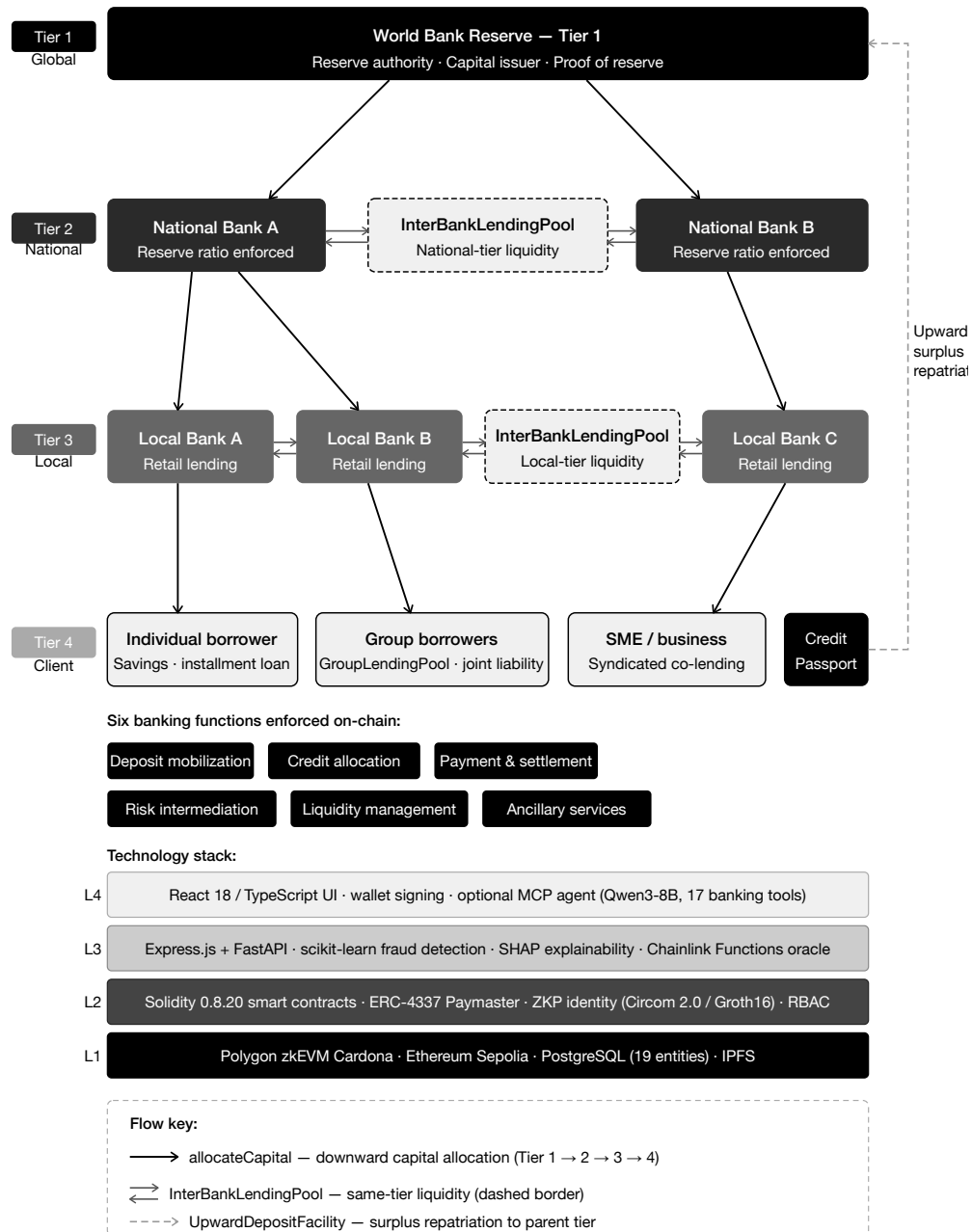


Figure 1.1: Crypto World Bank system overview

1.1 Background

Local lenders, or our target users, for this aspect of lending through layered institutions—supranational bodies that is provided by development finance channels have structural costs. Idle capital is usually locked by correspondent banking in pre-funded nostro/vostro accounts [114], cross-border transactions that take around two to five days cost roughly \$42 per transaction [19], charges for remittance corridors average 6.49%, which is above the UN SDG target of 3% [20]. Due to unequal access to traditional banking systems, 1.4 billion adults remain unbanked [9] and MSMEs face a \$4.5 trillion annual financing gap [15].

In decentralized financing, lending, collateral, and interest can be automated by enabling rules on-chain transparently. However, institutions like Aave v3 that holds \$26.3 billion TVL [21] and the broader DeFi lending market exceeds \$55 billion [8], use flat pool design that ignore institutional hierarchy, inter-tier exchange, and accessing clients via local bank lend path which is more generalized for normal people with less knowledge about DeFi.

Contribution of smart-contract and blockchain. Smart-contracts can develop lending rules as immutable, atomic, with auditable on-chain logic. This technology replaces the bilateral ledger reconciliation with a system of shared record, when a local bank initiates a loan, the World Bank's reserve ratio updates on the same transaction status [19]. DeFi extends this feature to financial services without licensed custodians. Here in *The Crypto World Bank* adds three dimensional feature that is missing from the existing institutional protocols: institutional hierarchy (four-tier capital flows), multi-entity operations (solidarity groups, syndicated loans, tranching pools), and a connection to the evolving development finance framework identified as absent in the literature [1].

Optional conversational interface. Every user interacts with this system primarily through the React UI and wallet credential signing. For development purposes and demonstration, a locally hosted agent (Qwen3-8B + MCP, 17 banking tools) provides an alternate path to get the essential tasks done, including all the customer services that require system interaction information guidance and assistance in acquiring a loan or exchange finances with the system. In the final build it is planned to be replaced with secured API-based services.

Technical foundations. Consensus. The chain is an append-only ledger: nodes agree on history, and no one can rewrite past blocks without controlling most of the network [63].

EVM execution [50]. Loan logic runs as deterministic bytecode on Ethereum-compatible chains. Storage writes are the main cost driver (Section 5.1).

State machines. Each loan moves through clear stages—PENDING → APPROVED → ACTIVE → REPAYING → COMPLETED/DEFAULTED—with only authorised roles allowed to advance it.

Oracles [55, 95]. Contracts cannot read off-chain data on their own. ML risk scores reach the chain via a commit-reveal oracle (Section 3.3); Chainlink Feeds, Automation, and Proof of Reserve handle prices, due-date checks, and reserve attestation. Live ML wiring starts after pre-thesis 2.

Polygon zkEVM. We deploy on Cardona testnet (Section 3.2) for low fees and L1-backed validity proofs.

1.2 Rationale and Motivation

Real development finance depends on layered institutions, yet settlement friction, opaque reserves, and correspondent-banking idle capital leave both intermediaries and unbanked clients underserved [20, 114]. Flat DeFi protocols automate lending at scale but ignore institutional hierarchy, cross-tier flows, and the Local-Bank-to-client path that microfinance and development banks use in practice [1]. This study is motivated by the opportunity to encode that hierarchy on a programmable ledger—with enforced reserves, role-based governance, and auditable capital movement—while supporting (not replacing) human credit judgment through lightweight analytics and an optional conversational interface. The thesis specifies a reusable architecture; it does not claim a particular national deployment target.

1.3 Problem Statement

Three primary forces of motivation for this study: development-finance has restricting transparency and settlement; the absence of multi-tiered banking functions in DeFi (deposits, solidarity and lending); the keen interest in combining blockchain auditability with ML computational analytics. There are six structural inefficiencies noticeable in development finance routes for individual borrowers from national and local banks:

1. **Lack of transparency:** there is very little chance of the lower tier users to verify capital management, reserve levels or lending decisions are made by the higher tiers, very less transparency on how money flows through the different levels of systems. Only the top tiers get the information and records of reserves and audited at most quarterly.
2. **Sluggish settlements and idle capital:** cross-border transactions are quite expensive, average \$42 per transaction [19], and time consuming taking two to five days [114].
3. **Inconsistent risk evaluation:** fraud and credit assessment are completely relied on human judgement, borrowers are generally re-evaluated at each tier or institutions based on information available at that tier, no unified credit history.
4. **Access and trust barriers:** the cost of inter-institutional trust is quite expensive for legal and audit process. It creates congestions in developing countries where investors earn returns roughly 3% below compared to developed markets [22].
5. **No programmable savings:** depositors in developing economies have no real-time access to visibility into how savings are processed and deployed.
6. **No on-chain group credit:** without individual collateral, borrowers cannot access programmable on-chain mutual group lending with shared responsibility. Real-world organizations like BRAC exist but does not function like DeFi.

DeFi enabled automatic lending transparency at scale such as Aave v3 with \$26.3 billion

TVL, \$17.7 billion borrows [21], Compound v3 (\$1.4 billion) [23], MakerDAO with \$10.5 billion stablecoin issuance [24]. Despite large market coverage they exhibit four limitations for institutional development finance:

- **Flat lending models.** Aave/Compound uses one protocol for many asset pools (USDC, ETH, etc.). There is no cross-tier allocation, no institutional hierarchy, no difference in role-based governance. Here our CWB models institutional relationships that go deep into what kind of services to provide to what kind of roles/institutions to increase liquidity in overall lending and exchange situations.
- **No AI risk analytics.** DeFi relies on collateral ratio, utilization curves. There is no behavioral fraud detection or explainable credit support system.
- **No tiered governance.** DeFi protocol treats tokens as a measurement of voting, not based on what is the role of the client or user or institution in the bigger to smaller picture as it lacks relation to a deeper level like institutions.
- **Single-level institutional DeFi.** Maple & Goldfinch [19–20] with \$2.6–3.8 billion TVL and \$680 million originations in 18+ countries respectively serve institutions, but none models a full multi-tier banking system with interbank flows, lack hierarchical capital flows.

1.4 Objectives

The objectives of this project are:

1. Design and specify all features and constraints of a four-tier lending architecture on an EVM-compatible blockchain and complete a full testnet deployment (World Bank → National Bank → Local Bank → Client). The whole development cycle is planned for Phases I–IV.
2. Design not only pool lending, but specify an extended banking product suite on the hierarchical system (deposits, savings, and solidarity group lending).
3. Investigate a lightweight off-chain analytics layer using ML, training and testing the models in practical testnet for fraud detection, anomaly detection, explainable review, and an optional MCP-based conversational interactive agent for ease of use.
4. Specify programmable on-chain lending with borrowing limits based on financial behaviour analysis, installments, and role-based access control and information via smart contracts.
5. Plan and execute testnet deployment using Polygon zkEVM Cardona, Ethereum Sepolia, and evaluate hierarchical controls in the final thesis phase.
6. Document governance for future improvements, security analysis and feature improvements, role-based regulatory considerations, and a controlled rollout toward sandbox piloting.

These objectives are tested through five research questions: **RQ1:** Does a hierarchical blockchain lending architecture reflect development-finance capital flows more faithfully than flat DeFi protocols? **RQ2:** Can on-chain transparency, reserve controls, and role-based governance reduce opacity and friction across tiers? **RQ3:** Does a lightweight off-chain analytics layer offer practical, auditable fraud and lending support? **RQ4:** Can the architecture deploy on public testnets with the RBAC, capital-flow, and reserve properties required for institutional banking? *Scope: prototype viability only—not production readiness or regulatory compliance.* **RQ5:** Can deposit-funded and solidarity group lending be represented as programmable, auditable on-chain mechanisms feasible for retail users?

1.5 Research Contribution

Four things this thesis actually adds—each with honest limits on what is built vs. specified. **C1—Four-tier lending architecture:** Most DeFi lending is flat; we model development-finance-style capital flow across four tiers with reserve checks and on-chain roles for World, National, Local, and retail clients [1, 37, 90] (*built now:* contract interfaces and workflow spec; *later:* testnet deploy, Phases I–II). **C2—Group lending on-chain:** Solidarity-style group loans with shared consent, split installments, and over-borrow caps are specified as EVM contracts (Section 3.5; PoC in Phase II). **C3—Explainable ML through an oracle:** Fraud and anomaly scores leave the ML service, get explained with SHAP, and land on-chain before an approver can sign (Section 3.3; training after pre-thesis 2). **C4—Privacy-preserving KYC path:** Prove eligibility with zk-SNARKs instead of posting identity on-chain—KYC circuit specified now; fuller AML circuit left for Future Work. We are not pitching a bank killer: institutions care about legal enforceability, not just clever contracts [65]; CWB is a composable layer for development finance and MFIs (Section 1.8), not a replacement for correspondent banking overnight. Section 1.9 maps where each topic introduced in this chapter is expanded across Chapters 2–6 and the appendices.

1.6 Proposed Solution

The Crypto World Bank will implement a four-tier on-chain lending model: Tier 1 World Bank reserve custodian → Tier 2 National Banks → Tier 3 Local Banks processing retail loans → Tier 4 clients building on-chain credit history (Figure 1.1), with six standard banking functions enforced on-chain—deposit mobilization, credit allocation, payment and settlement, risk intermediation, liquidity management, and ancillary services—rather than discretionary audit alone. Banks also borrow from peers, park surplus with parent tiers, and syndicate large loans; the platform replicates four-tier downward flow, same-tier pools, upward deposits, and multi-bank co-funding on-chain (Figure 1.2).

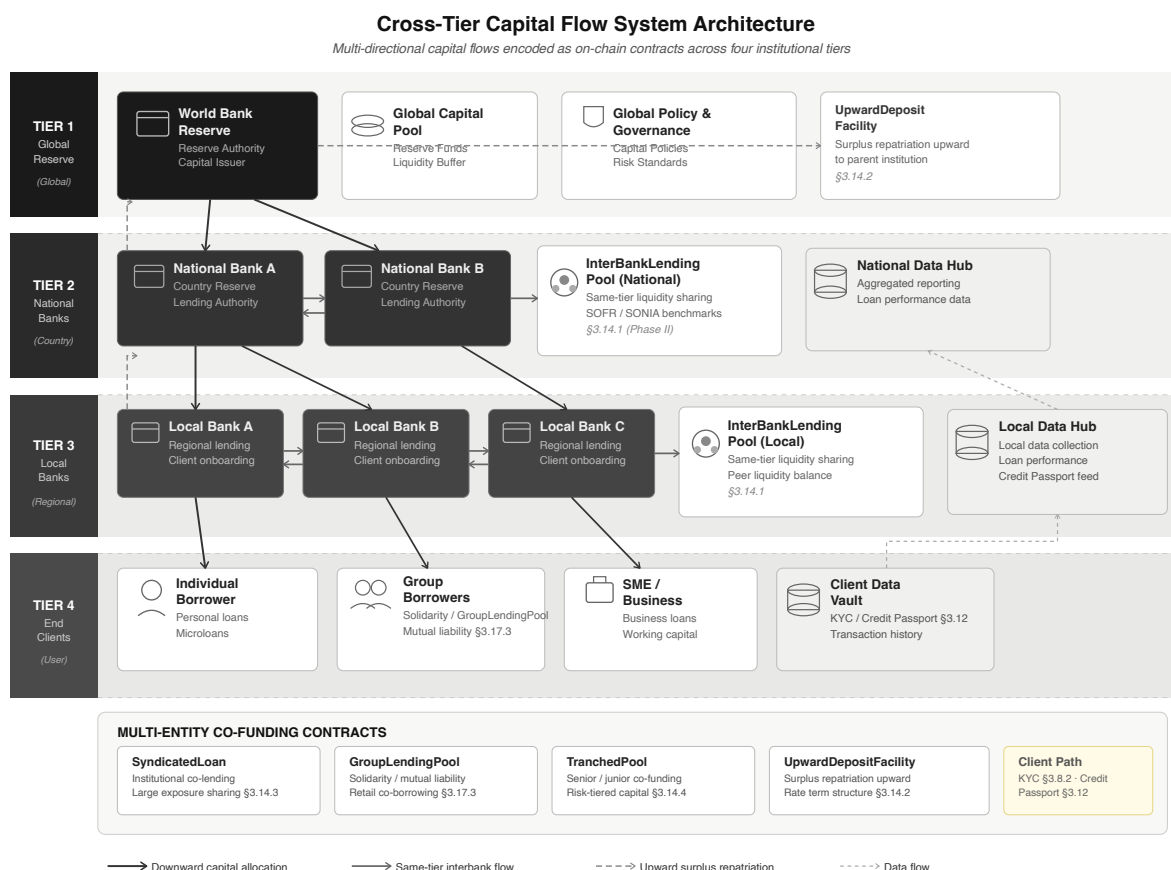


Figure 1.2: Cross-tier lending flow directions

Can a group of entities lend or fund together? Syndicated loans, solidarity groups, tranching pools, and upward surplus deposits (Section 3.5) coexist with same-tier peer liquidity pools (Section 3.5); retail clients enter through Local Banks via KYC, Credit Passport, and borrowing limits, with syndication reserved for large loans.

1.7 Methodology in Brief

There are four Agile phases to this methodology for this project. This report is designed with the architecture specification. The development phase will start from Phase II after pre-thesis 1. In this report there is architecture, contracts, database, and ML pipeline. The Phase II development phase will demonstrate the loan cycle. Phase III will start the implementation of ML scoring and the Chainlink oracle (Section 3.3). Phase IV will show the evaluation, metrics, and the overall feasibility of the project in the real world. Stack: Solidity, React, Express/FastAPI, scikit-learn on Polygon zkEVM Cardona. There is also an AI chat agent that will be built as an optional feature that will run alongside the project to help the clients navigate better with the system.

1.8 Scopes and Challenges

In scope now. In our development phases, we have calculated that we can implement a limited amount of our full vision: the four-tier lending architecture, loan lifecycle workflow, borrowing limits for the clients and the banks, the design of Oracle fraud analytics, and the training and the live demonstration of the AI/ML workflow. It can be shown to a limited amount on the prototype builds only.

Out of scope for now. Mainnet deployment, live retail payments in the real world, production KYC/AML at the banking level and client interaction, stress testing the whole project, and providing an MVT checklist before the product is ready is not quite possible yet.

Main risks. There are a couple of risks that we have noticed for our project, starting with the fraud labels may not be completely accurate based on the limited amount of data we have for our system architecture. We need more synthetic data that will be collected over time. If regulations have changed—mitigated by testnet-only builds—then we might have to change the architectural elements. The gas toll on micro-loans may not be sustainable. The ML layer with SHAP may keep borrowers in the loop in the system since it is not 100% accurate and accuracy of the system depends on the training process. Rest of the economics and trade-offs are discussed in Chapter 5.

Economic sustainability. On Cardona, 10–15 txs on a typical loan cost under \$0.15—small next to interest earned. Mainnet micro-loans would need tighter gas work.

Institutional capture. Another huge issue that needs to be addressed is that Tier 1 and the National Banks, which have huge reserves and influence over the rates, reserves, and who gets credit, might influence the overall system and cost-to-performance ratio. It might temper the trust and relationship with the retail customers and smaller banks.

Stablecoin-first lending. Retail clients are not to be carrying high-priced positions like ETH, since the prices of Ethereum might fluctuate. MiCA and the U.S. GENIUS Act support the idea that stablecoins should be more generalized for authorized cryptocurrency. There are many settlement projects like *mBridge* and *Agora* [103, 110], moving the value of cryptocurrencies across borders, but it is not a replacement for our lending system. The architectural system for stablecoin pools and L2 deployment (Section 3.5) gives the liquidity move fluidly from the top tier to the bottom, and the surplus to go up from the bottom.

1.9 Thesis Organization

- **Ch. 1 — Introduction:** problem, objectives, approach (CO1).
- **Ch. 2 — Literature:** PRISMA review and protocol comparison (CO9).

- **Ch. 3 — Architecture:** tiers, contracts, data model, identity, products, security (CO2–4, CO12).
- **Ch. 4 — Methodology:** build phases, MVT scope, ML oracle, optional agent (CO5, CO10).
- **Ch. 5 — Evaluation:** feasibility, benchmarks, limits (CO6–7, CO11).
- **Ch. 6 — Conclusion:** contributions and future work; appendices hold contracts and deployment notes.

Chapters 3–5 already carry the detailed architecture, methodology, and feasibility analysis for this pre-thesis 1 submission; richer implementation evidence, measured benchmarks, and updated evaluation will be added to those chapters after pre-thesis 1, in the final thesis stage. Hard numbers—testnet deploy, ML benchmarks, formal proofs—are scheduled for pre-thesis 2.

Chapter 2

Literature Review

2.1 Preliminaries

Core terms—DeFi, smart contracts, over-collateralization, XAI, PoS, CBDC, solidarity/group lending, ALM, flash loans, ZKP, and Health Factor—are defined in the **Glossary** appendix (Appendix D in the final thesis assembly; the current pre-thesis build temporarily carries the WorldBankReserve interface as Appendix D). Platform-specific terms (*Platform Solvency Ratio*, *Provisional Credit Tier*, *Cold-Start Credit Pathway*) appear there as well. Readers may skip to Section 2.2 if already familiar with blockchain fundamentals (Section 1.1).

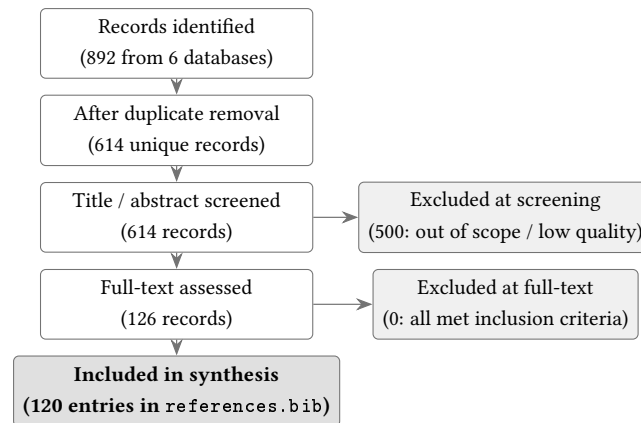
2.1.1 Review Methodology (PRISMA-Style Frame)

PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is a standard way to search, screen, and report which studies made it into a review—so readers can see what was included, what was dropped, and why. We use a PRISMA-*style* frame here: the same transparent screening steps, applied to design literature rather than a clinical meta-analysis [66].

We followed that frame, not a narrative summary alone. Sources had to be peer-reviewed or institutional (2017–2026), relevant to at least one CWB topic (lending, ML fraud, identity, microfinance, security, regulation, or payments), and clear enough to reuse (metrics, reserve rules, gas figures, F1 scores).

We searched ACM, IEEE, Elsevier, MDPI, arXiv, and BIS/World Bank with keywords around blockchain, DeFi, lending, fraud, ZKP, and development finance.

Table 2.1 ranks the top ten; Figure 2.1 shows screening counts. PRISMA-based banking reviews [17] use the same standard.

**Figure 2.1:** PRISMA-style literature screening flow**Table 2.1:** Top-10 ranked literature sources

#	Source / Module / Operationalized In	Headline Finding
1	Werner et al. 2022 [1] · DeFi mechanics · Ch. 3	DeFi lending is uniformly flat, pool-based, overcollateralized
2	Tan 2023 (IMF) [5] · CBDC & hierarchy · Ch. 3	Two-tier CBDC distribution (central bank → commercial banks → users); hierarchical infrastructure is the key design choice
3	Palaiokrassas et al. 2024 [3] · Fraud detection · Sec. 3.3	Multi-chain ML on 54M tx, F1 0.76–0.85 with behavioral features
4	Adom et al. 2022 [4] · Explainable AI · Sec. 3.3	SHAP outperforms LIME on lending datasets
5	Weber et al. 2019 / [85] · Elliptic Graph fraud · Sec. 4.2	Graph-structured fraud signal in BTC transactions
6	McMahan et al. 2017 / FedAvg [68] · Federated learning · Sec. 4.2	Privacy-preserving model averaging across institutions
7	BIS WP 905 [61] · Stablecoin risk · Secs. 1.8, 5.4.4	Algorithmic stablecoins unsuitable for EM DeFi; USDC/USDT recommended
8	Atzei et al. 2017 [7] · Smart contract security · Sec. 3.6	SoK of EVM attack vectors
9	EU MiCA + GENIUS Act [75, 77] · Regulation · Sec. 5.4.4	Stablecoin issuer authorization, reserve disclosure
10	Bangladesh Bank SP2025-02 [80] · MFI precedent · Sec. 3.5	Female-to-male account parity 49%/49% in 2025; 89% BRAC loans to women

2.2 Review of Existing Research

This part of the paper explores all the literature chosen through a PRISMA-style screening flow to select the papers that best fit our project. We surveyed existing work related to DeFi lending, ML fraud detection and explainable AI, hierarchical development finance and cross-tier finance flows, group or microfinance, smart contracts and gas economics, retail banking and payments, remittance, monetary-policy distributional effects, real-world asset tokenization, stablecoin regulation and institutional settlement, and how inference can be achieved in systems like ours.

2.2.1 Decentralized Lending and DeFi Protocol Design

This is one of the most important parts of the paper: how decentralized financing works for lending, and how DeFi protocols are designed as public smart contracts. Utilization curves set interest rates; oracles trigger liquidations on collateral; settlement stays atomic across currencies. The model’s real strength is real-time auditability—per-block interest and transparent on-chain state. It also comes with structural limits for development finance: capital still flows one way through flat pools, with no inter-tier allocation or multi-entity co-funding—a crucial gap for our project.

Werner et al. [1] summarize DeFi as homogeneous, pool-based, and over-collateralized, with no institutional hierarchy. Bastankhah et al. [2] show adaptive fast/slow rate control outperforming static Aave/Compound curves [51], which informs our kinked-rate model (Section 3.5). Chen et al. [39] test multi-chain lending layouts to handle higher throughput. Yang and Cui [40] propose practical metrics—utilization stability, liquidation efficiency, and rate responsiveness—for judging how well a lending stack performs. Hartmann and Hasan [41] look at gas cost and digital identity in peer-to-peer lending. Dao et al. [52] map traditional credit scoring onto DeFi borrowing limits.

2.2.2 Machine Learning, Explainable AI, and Extended AI Security

Palaiokrassas et al. [3] achieve F1 0.76–0.85 on 54M multi-chain DeFi transactions using behavioral features versus 0.08 with raw transaction data alone—direct input to Random Forest feature engineering. Liu et al. [6] supply Isolation Forest for unsupervised detection when labeled fraud is scarce. Hasan et al. [42] train fraud classifiers via federated learning on encrypted features, informing cross-bank intelligence without raw data sharing.

Adom et al. [4] find SHAP more consistent than LIME for loan explanations; Yuan [49] reaches AUC >0.92 with SHAP on credit-default models—motivating SHAP waterfall review before approvals. Beyond the core pipeline, graph features from Palaiokrassas et al. [3] support GNN ring detection; federated learning [42] suits National/Local Bank boundaries; ContractWard [43] and SoliAudit [44] enable real-time exploit monitoring; RL rate tuning and NLP income extraction remain future-work options.

2.2.3 Hierarchical Distribution, Cross-Tier Flows, and Group Lending

Tan [5] models two-tier CBDC distribution (central bank $\rightarrow$ commercial banks $\rightarrow$ users), paralleling the four-tier hierarchy. BIS [46] prefers two-level retail CBDC with interoperability constraints, informing the dual-currency facility. Asaduzzaman et al. [47] report up to 60% admin-cost reduction from smart-contract microcredit—literature precedent for `GroupLendingPool`, not a deployment-jurisdiction claim.

Policy literature establishes hierarchical distribution but sparse programmable *multi-directional* coordination. Kiff et al. [106] distinguish multi-tier CBDC with supervised intermediaries; Bindseil [105] analyses tiered remuneration for `UpwardDepositFacility` rate spreads (Section 3.5); Auer and Böhme [107] hybrid CBDC mirrors client-via-Local-Bank access with Tier 1 reserve authority. *Project Pine* [103] validates layered contract factories for reserve and collateral dependencies extended by `InterBankLendingPool` and `SyndicatedLoan`. Dong et al. [104] model cross-tier direct financing in supply chains—parallel to downward allocation to capital-constrained retail clients (Figure 1.2).

Grameen/BRAC solidarity groups achieve $>95\%$ repayment through social monitoring that on-chain mutual liability cannot fully replicate; `GroupLendingPool` encodes multi-signature consent and pool claims at pre-thesis stage. Güçük et al. [53] review privacy and trust in blockchain microlending.

2.2.4 Smart Contract Security and Layer 2 Scalability

Atzei et al. [7] catalogue reentrancy, access-control, and overflow vectors—motivating `OpenZeppelin` guards, Solidity 0.8.20, RBAC, Slither, Mythril, and Phase IV formal verification. Wang et al. [43] (`ContractWard`) reach F1 0.96/0.93 on bytecode vulnerability classes; Liao et al. [44] (`SoliAudit`) cut audit time $\approx 70\%$ via ML-prioritized fuzzing; So et al. [45] (`VERISMA`) report 98.4% precision with zero false positives on arithmetic safety—candidate for reserve-ratio invariants.

Kim and Kim [48] show batching and calldata compression save 30–50% mainnet gas, supporting Polygon zkEVM Cardona deployment where multi-step four-tier lifecycles multiply savings.

2.2.5 Correspondent Banking, Retail Payments, and Settlement Rails

BIS [16] and FSB [27] document nostro/vostro idle capital, MT103 routing, and 2–5-day settlement. World Bank remittance data [20] average 6.49% on a \$200 transfer (Sub-Saharan Africa $>8\%$). Cerutti et al. [116] and Du et al. [110] show digital and stablecoin rails can undercut legacy costs but depend on corridor off-ramp depth—scoped separately (Section 4.3).

Retail literature examines stablecoin remittance, merchant checkout, and on/off-ramps for post-MVT modules (`CurrentAccount`, `P2PTransfer`). Ante [108] finds 26% U.S.

remittance-user stablecoin adoption, supporting tiered KYC gating. Du et al. [110] document the fiat–stablecoin–fiat “sandwich” pattern. Li et al. [109] (**CLEAR** framework) find programmable settlement advantages but weak open-loop dispute recourse; Zhang et al. [112] warn on refund volatility; Fröhlich et al. [111] show on-ramp friction dominates POS UX. Lin et al. [113] (*SecurePay*) and BIS *Project Sela* [119] validate Local-Bank-sponsored multi-intermediary flows; Cimaszewski et al. [118] blueprint jurisdictional DLT settlement. CPMI [114], BIS Papers 170 [115], and FATF [117] define ramp and VASP compliance for platform-mediated P2PTransfer—not permissionless unhosted-wallet remittance. No prior work composes four-tier development-finance lending with closed-loop retail rails in one architecture.

2.2.6 Monetary Policy, Tokenization, and Institutional Landscape

Cross-country evidence links QE to wealth inequality [28]; Federal Reserve metro analysis [29] shows low-income harm from tightening—motivating algorithmic, transparent rate and reserve parameters rather than opaque committee discretion. Tier-uniform utilization rates reduce loan-officer price discrimination; the Cantillon effect does not apply where USDC is fully reserved and the platform creates no new money [30].

R3 Corda reports \$17B tokenized RWAs [31]; Centrifuge exceeds \$1B across eight chains [32]; World Bank FundsChain tracks disbursements on 13 projects scaling to 250 [33]—institutional precedent for hierarchical on-chain finance beyond fund tracking alone.

Sygnium [65] notes institutional allocators require legal enforceability (Aave Arc $\approx$ \$50k TVL). mBridge [78] and *Project Agora* [79] are settlement rails, not retail lending hierarchies; 94% of central banks research CBDCs [71]. CWB positions as a lending-layer specification compatible with these rails.

2.2.7 Graph ML, Federated Learning, Regulation, and Identity Primitives

DeFi fraud is relational: Cheng et al. [93], Chang et al. [93], and DeFiGuard [67] outperform flat ML on graph structure—motivating a GNN extension with hierarchical tier edges (Section 4.2). Federated learning [42, 97] and PrivChain-AI (94.7% accuracy) suit banks that cannot share raw records; FedAvg aggregation at National tier is architecturally natural (Section 4.2).

MiCA [75, 76] and the U.S. GENIUS Act [77] anchor stablecoin-first retail design (Section 1.8; mapping in Section 5.4.4). Buterin et al. [69] introduce non-transferable SBTs for portable credit passports [70] (Section 3.5).

2.2.8 Oracle-Mediated ML and Optional LLM Interfaces

Luo et al. [94] survey AI fraud detection across the DeFi lifecycle; Caldarelli [95] identifies ML-score oracle delivery as under-researched beyond price feeds. BCCC-DeFiFraudTrans-2025 [83] (1,026,867 labeled transactions) is the planned primary training corpus after pre-thesis 2 (Section 4.2); Fazliani et al. [96] and Alhasawi et al. [97] supply semi-supervised and federated baselines.

The optional conversational add-on uses an 8B open-weight LLM with QLoRA, RAG, and refusal layers. Finance LLM studies [72, 73, 74] report productivity gains but document regulatory-hallucination risks—motivating the red-team protocol in Section 4.3.

2.3 Literature Review Summary

Tables 2.2 and 2.4 consolidate the reviewed works by methodology, findings, and project relevance. Extended six-part synthesis tables are maintained in the thesis repository [17].

Table 2.2: Literature review summary (Part 1 of 2)

Author & Focus	Methodology	Key Findings	Relevance to Our Project
Palaiokrassas et al. (2024) [3] · DeFi fraud	XGBoost, NN on 54M+ txs	F1: 0.76–0.85 vs. 0.08 with features alone	Random Forest fraud modeling
Adom et al. (2022) [4] · XAI lending	LIME / SHAP comparison	SHAP deeper, more consistent than LIME	Primary explainability method
Tan (2023) [5] · CBDC hierarchy	Developing-nation model	Two-tier CBDC distribution; hierarchy decisive	Four-tier architecture
Liu et al. (2008) [6] · Anomaly detection	Isolation Forest	Recursive partitioning; shorter paths	Unsupervised wallet detection
Atzei et al. (2017) [7] · SC security	SoK of Ethereum attacks	Reentrancy, access-control gaps	Security primitives, formal verification
Werner et al. (2022) [1] · DeFi SoK	Survey of 12+ protocol types	Pool-based, overcollateralised; no hierarchy	Core gap CWB addresses
Bastankhah et al. (2024) [2] · Adaptive lending	Dual fast/slow control	Dynamic rates beat static curves	Kinked-rate validation
Hartmann & Hasan (2021) [41] · P2P lending	Smart-contract origination	Gas minimisation; no intermediary	Four-tier P2P design
Hasan et al. (2022) [42] · Federated fraud	FL on encrypted features	Comparable to centralised ML	Cross-bank intelligence
Wang et al. (2020) [43] · SC vulnerabilities	ML on bytecode	F1: 0.96 access control; 0.93 leakage	Security verification models
Liao et al. (2019) [44] · Audit automation	Classification + fuzzing	70% faster than manual audit	Security audit strategy

Table 2.4: Literature review summary (Part 2 of 2)

Author & Focus	Methodology	Key Findings	Relevance to Our Project
BIS et al. (2020) [46] · CBDC design	Retail distribution analysis	Two-tier preferred; interoperability gaps	Dual-currency facility
Kiff et al. (2020) [106] · CBDC models	IMF architecture survey	Multi-tier via supervised intermediaries	Wholesale-to-retail path
Bindseil (2020) [105] · CBDC rates	ECB tiered remuneration	Preserves bank intermediation	Asymmetric rate spread
Dong et al. (2023) [104] · Supply-chain finance	Three-tier game model	Cross-tier direct financing	Downward client lending
Project Pine (2025) [103] · Hierarchical SCs	NY Fed / BIS prototype	Factory-to-facility hierarchy	IBLP / SyndicatedLoan
BIS/FSB [16] · Cross-border settlement	Nostro/vostro cost analysis	2–5 days; \$42/transfer avg.	Near-instant four-tier settlement
Beyer et al. (2025) [28] · QE inequality	Cross-country 1999–2019	QE raises wealth inequality	Algorithmic monetary parameters
World Bank (2025) [33] · Dev. finance blockchain	13-project pilot	Reduces reporting burden	Multi-chain deployment
Chen et al. (2024) [39] · Multi-chain lending	Cross-chain design	Higher throughput, less congestion	Multi-chain strategy
Yang & Cui (2023) [40] · Protocol evaluation	Quantitative risk metrics	Utilisation, liquidation, responsiveness	Tier benchmarks
Asaduzzaman et al. (2020) [47] · Smart micro-credit	Smart-contract MFI	60% admin overhead reduction	GroupLendingPool
Kim & Kim (2024) [48] · Gas optimisation	Batching, calldata compression	30–50% gas savings	L2 deployment
Yuan (2024) [49] · SHAP credit default	RF, GBM + SHAP	AUC >0.92, interpretable	Per-client risk explanation

Table 2.6: Literature review summary (Part 2 of 2, continued)

Author & Focus	Methodology	Key Findings	Relevance to Our Project
Ante (2025) [108] · Stablecoin remittance	U.S. survey (n=866)	26% adoption; literacy drives uptake	KYC ladder, P2PTransfer
Du et al. (2026) [110] · Payment rails	Corridor cost comparison	Stablecoin sandwich undercuts banks	PSP on/off-ramp work
Li et al. (2026) [109] · Retail payments SoK	CLEAR framework	Closed-loop advantage; weak disputes	Merchant checkout design
Zhang et al. (2024) [112] · Stablecoin checkout	Operations model	Refund volatility disutility	FXModule, auto-conversion
Fröhlich et al. (2022) [111] · Crypto POS	Lightning usability	On-ramp friction dominates	Fiat-funded CurrentAccount
Lin et al. (2025) [113] · Platform payments	Permissioned chain prototype	≈256 TPS; multi-intermediary escrow	Local-Bank merchant flows
CPMI / BIS [114, 115] · Stablecoin policy	Cross-border analysis	Ramps gate acceptance	Ramp integration
FATF (2021) [117] · VASP / P2P	Risk-based AML	Platform transfers need originator data	Sponsored P2PTransfer

2.4 Comparative Protocol Analysis

No surveyed protocol combines multi-tier hierarchy, oracle-mediated explainable ML, and compliance-aware identity. Table 2.7 compares CWB against Aave, Compound, Maker, Maple, and Goldfinch on eleven architectural dimensions.

Feature	Aave	Comp.	Maker	Maple	Gfinch	CWB
Institutional hierarchy	✗	✗	✗	✗	✗	●
Cross-tier capital flow	✗	✗	✗	✗	✗	⌚
Same-tier interbank lending	✗	✗	✗	✗	✗	⌚
Syndicated / co-lending	✗	✗	✗	Part.	Part.	⌚
Upward capital repatriation	✗	✗	✗	✗	✗	⌚
Solidarity group lending	✗	✗	✗	✗	Part.	●
AI/ML fraud detection	✗	✗	✗	Man.	Man.	●
SHAP explainability	✗	✗	✗	✗	✗	●
ZKP KYC compliance	✗	✗	✗	✗	✗	●
Kinked interest rate curve	✓	✓	Gov.	✓	✗	⌚
Role-based access control	Part.	Part.	Gov.	✓	Part.	●
Institutional / L2 focus	✗	✗	✗	✓	Part.	●
TVL / Status (2026)	\$26B	\$1.4B	\$10B	\$2.6B	\$680M	Planning

Table 2.7: Protocol comparison on eleven architectural dimensions

This answers RQ1: flat-pool protocols differentiate assets, not institutional relationships or inter-bank flows; Maple, Goldfinch, and Morpho V2 [117, 90] lack multilateral development-finance capital allocation. CWB’s compound novelty (Section 1.5) spans hierarchy, explainable ML governance, and group lending. Tiers 1–3 use bespoke RBAC (Section 3.6) functionally equivalent to ERC-3643 token-level KYC gating [82]; enforced per-tier reserve ratios address opaque custody failures such as FTX’s 2022 collapse.

Six design decisions follow from the survey:

- Gudgeon et al. [56] → kinked-rate model (Section 3.5)
- Piper et al. [58] → ZKP identity path (Section 3.5)
- Tan [5], BIS [46], Kiff [106], Auer and Böhme [107] → four-tier capital flows
- Bindseil [105], Project Pine [103], Dong et al. [104] → multi-directional lending (Section 1.6)

- Asaduzzaman et al. [47] → GroupLendingPool
- Ante [108], Du et al. [110], Li et al. [109], Lin et al. [113], CPMI [114], FATF [117] → post-MVT retail stack (Section 4.3)

2.5 Summary of Key Findings

The literature review yields the following design-relevant conclusions:

- **Flat DeFi vs. tiered finance.** Pool-based protocols exceed \$55B TVL [8] but lack institutional hierarchy; CBDC research confirms tiered distribution as a workable pattern [5]. No prior work models four-tier reserve-enforced lending.
- **Settlement and retail rails.** Correspondent banking traps idle capital [18], settles in 2–5 days at $\approx \$42/\text{transfer}$ [19], and costs remittance users \$48–56B annually [20]. Stablecoin rails can compress costs [116, 110] but need ramp infrastructure; retail studies [109, 112, 111, 113, 119] favour Local-Bank-sponsored closed-loop flows over open permissionless checkout.
- **ML, XAI, and security.** Behavioral DeFi features lift fraud F1 to 0.76–0.85 [3]; Isolation Forest [6] covers unlabeled cases; SHAP beats LIME for regulatory explainability [4]. Contract vulnerabilities are well catalogued [7]; ML audit tools reach F1 > 0.93 [43], cut review time 70% [44], and formal verifiers such as VERISMART reach 98.4% precision [45].
- **Hierarchy, groups, and institutions.** Cross-tier CBDC and Project Pine literature [106, 105, 107, 103, 104] supports multi-directional programmable flows; analogue MFIs report $> 95\%$ group repayment [47] with on-chain mutual liability still an open gap. FundsChain [33], Corda RWAs [31], and Kinexys [34] show rising institutional blockchain adoption.
- **Scalability, privacy, and regulation.** Multi-chain lending [39] and L2 gas optimization [48] cut costs 30–50%; federated learning [42, 97] enables cross-bank fraud intelligence without raw data sharing. MiCA and GENIUS [75, 77] anchor stablecoin-first design; algorithmic parameters mitigate distributional opacity from discretionary monetary policy [28, 29].
- **Deposit-funded sustainability.** Sustainable MFIs rely on mobilized deposits, not donor capital alone—motivating savings vaults at the Local Bank tier alongside GroupLendingPool.

These findings justify the four-tier architecture, cross-tier lending modules, Random Forest / Isolation Forest / SHAP analytics, governance structure, and phased security verification pipeline developed in later chapters.

Chapter 3

System Architecture and Design

This chapter compresses Chapter 3 of `Pre-thesis_v31_final.tex` to the architecture an examiner needs at pre-thesis 1: scope, tiers, contracts, data partition, oracle path, and security framing. All architecture diagrams from the full thesis are included in Section 3.7.

3.1 Prototype Scope and Stack

Four Agile phases (I–IV) deliver a four-tier EVM lending stack on **Polygon zkEVM Cardona** with **Solidity 0.8.20**, **React + Wagmi** frontend, **PostgreSQL + FastAPI** off-chain services, and **Chainlink** for prices, automation, ML-score delivery, and proof-of-reserve. Table 3.1 lists build status; most on-chain modules are *specified* now and deploy in Phases II–IV.

Table 3.1: Prototype scope (summary)

Feature	Status
Four-tier RBAC (World / National / Local / Client)	Specified (Phase I)
Loan lifecycle + installments	Specified / Phase II
SavingsVault, GroupLendingPool, IBLP	Specified / Phase II
RF + IF + SHAP oracle pipeline	Planned (Phase III)
ZKP KYC, testnet deploy, Foundry sim	Planned (Phase IV)
Optional MCP agent (17 tools)	Optional (Phase III–IV)

3.2 High-Level Architecture

Four layers: (1) React wallet UI; (2) Express/FastAPI APIs and ML scorer; (3) 15 modular Solidity contracts across Tier 1–3; (4) Polygon zkEVM + Chainlink oracles. Tier 1 `WorldBankReserve` custodies reserves and allocates downward; Tier 2 `NationalBank`; Tier 3 `LocalBank` runs retail loans, KYC gates, and installments; Tier 4 clients hold Credit Passport SBTs and borrowing limits.

Core contracts (Phase I–II): `WorldBankReserve`, `NationalBank`, `LocalBank`, `SavingsVault`, `FixedDeposit`, `GroupLendingPool`, `FXModule`, `InsuranceFund`, `CurrentAccount`, `InterBankLendingPool`, `UpwardDepositFacility`, `TreasurySwap`.
Phase III stretch: `SyndicatedLoan`, `TranchedPool`, `NettingEngine`.

3.3 Oracle and ML Integration

Off-chain Random Forest + Isolation Forest produce a composite risk score s ; SHAP explains declines. Scores reach LoanController via **Chainlink Functions** (primary) or a **commit-reveal** relay (prototype). Approval is blocked until SCORE_REVEALED. Chainlink Automation triggers overdue installments; Price Feeds support FXModule.

3.4 Data Model and Partitioning

Twenty PostgreSQL entities (3NF) mirror on-chain loan state, installments, market data, ML audit logs, and optional agent sessions. **On-chain:** balances, roles, loan state machine, reserve ratios, committed scores. **Off-chain:** PII, SHAP JSON, chat history, indexed events. Identity uses zk-SNARK KYC (specified) + non-transferable Credit Passport SBT.

3.5 Cross-Tier Operations and Products

Capital moves **downward** (allocation), **same-tier** (interbank pools), and **upward** (surplus deposits). Group lending encodes solidarity consent; syndication handles large co-funded loans. Retail products: savings, fixed deposits, installment loans, FX display, insurance buffer.

3.6 System Modeling and Security

Use-case, activity, DFD, sequence, and capital-flow diagrams model nine actor types and the USDC loan lifecycle. Security follows five layers: wallet/RBAC, contract guards (OpenZeppelin), oracle trust boundaries, API hardening (EIP-712), and operational monitoring. Threat model maps reentrancy, oracle manipulation, and privilege escalation to controls.

3.7 Architecture Diagrams (Full Set)

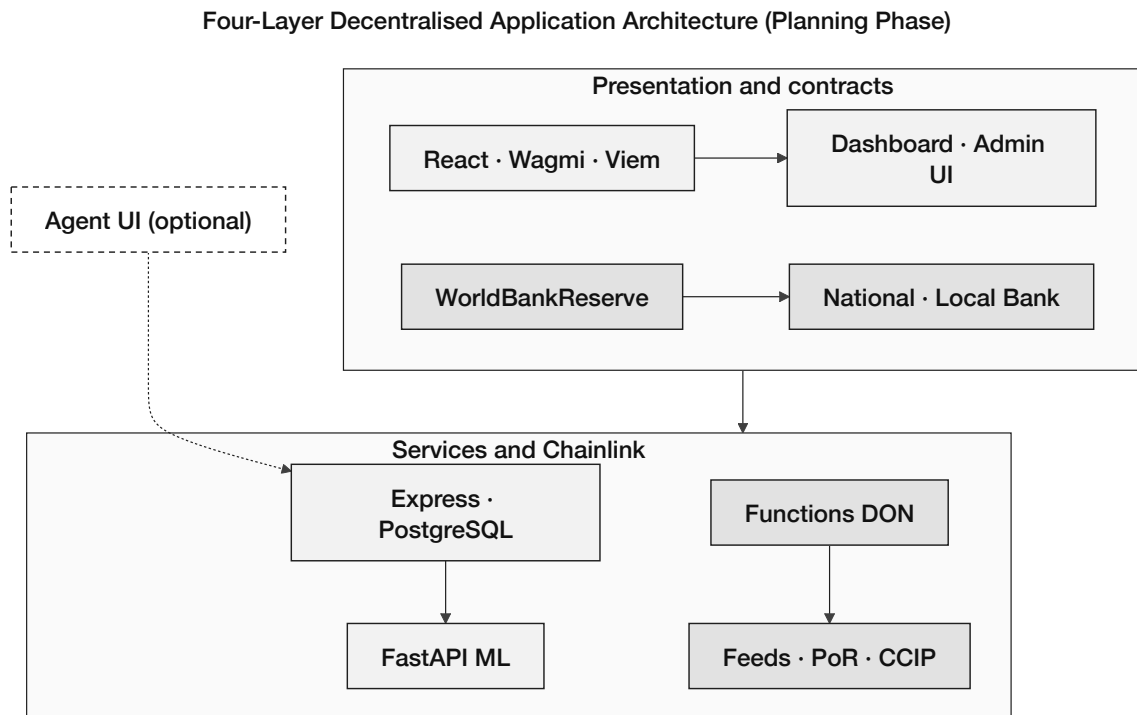


Figure 3.1: Four-layer application architecture

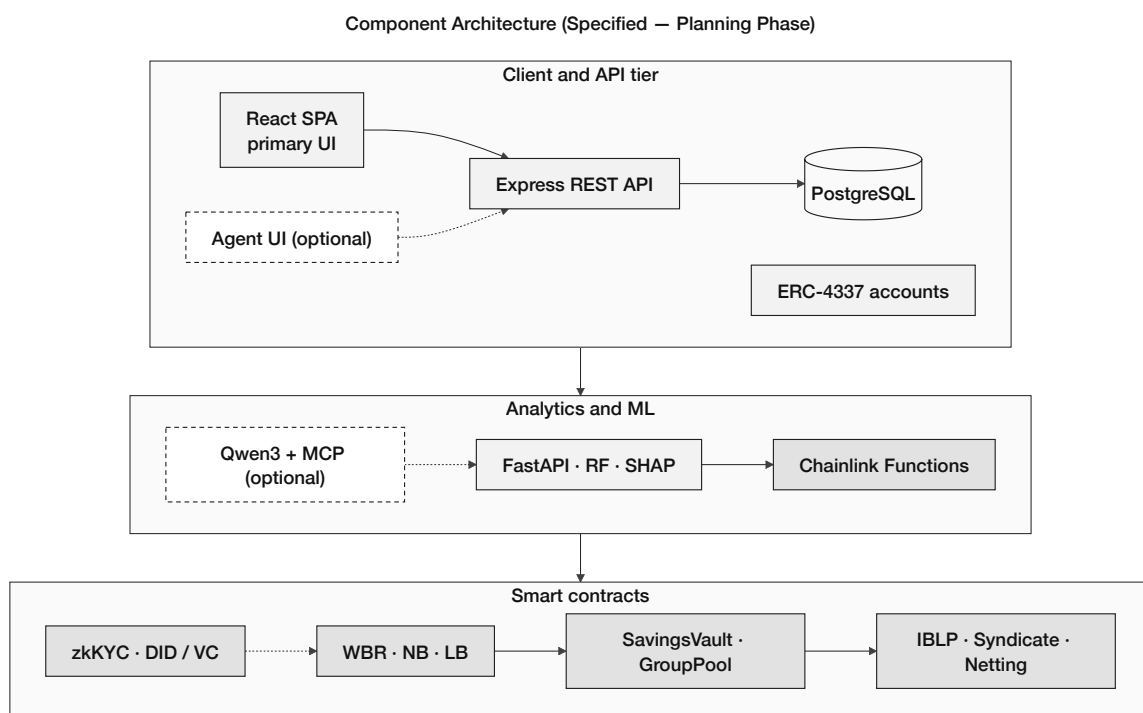


Figure 3.2: Component diagram (planning phase)

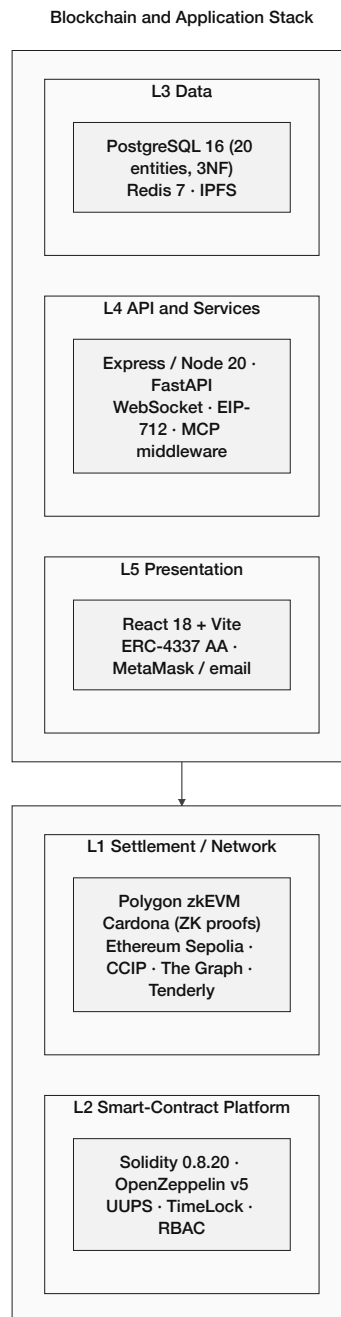
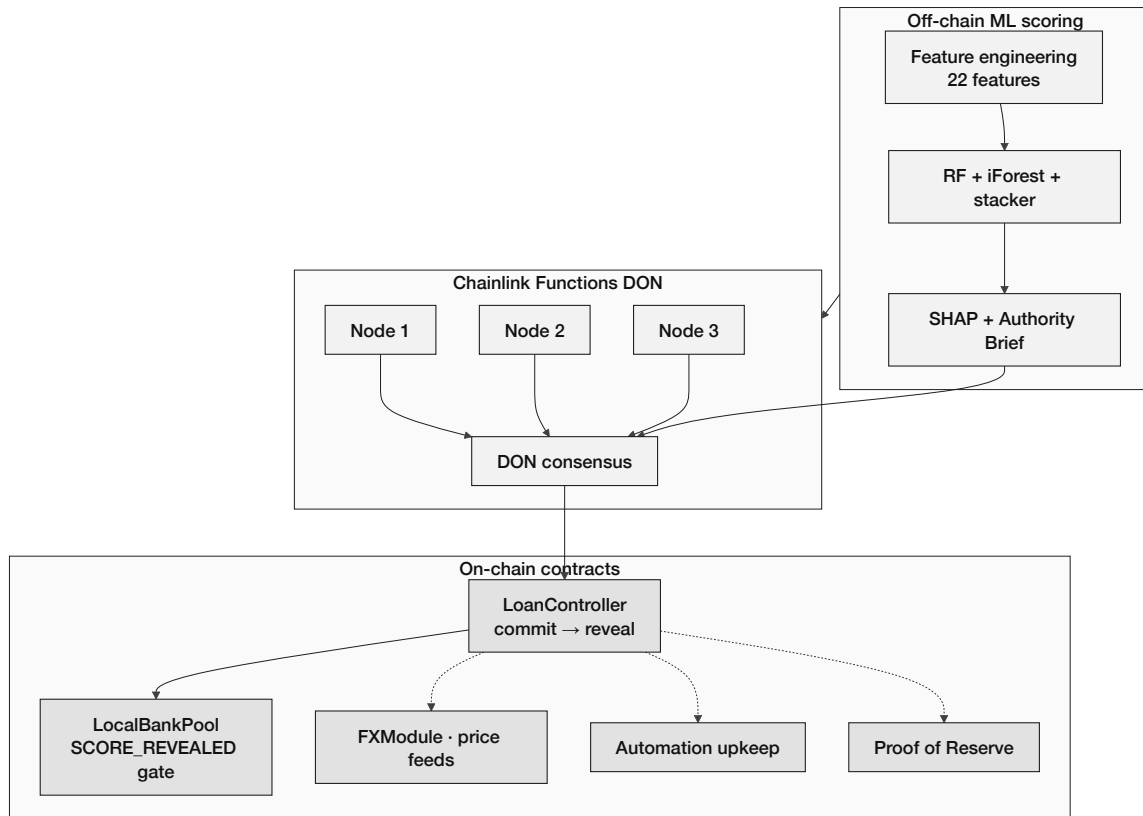


Figure 3.3: Layered blockchain and application stack

Chainlink Oracle Architecture - Off-Chain AI to On-Chain Decision

**Figure 3.4:** Chainlink oracle architecture

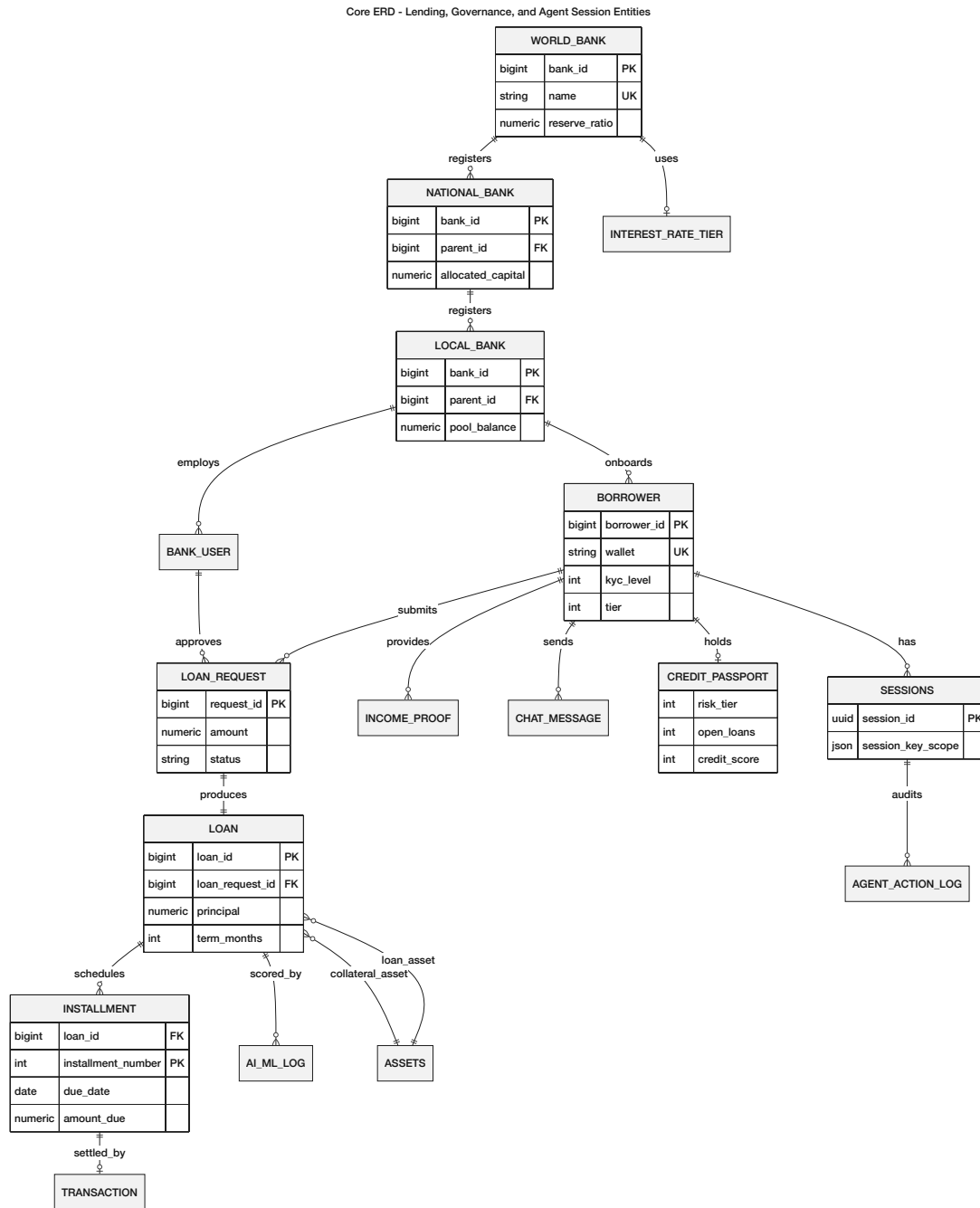


Figure 3.5: Core system entity graph

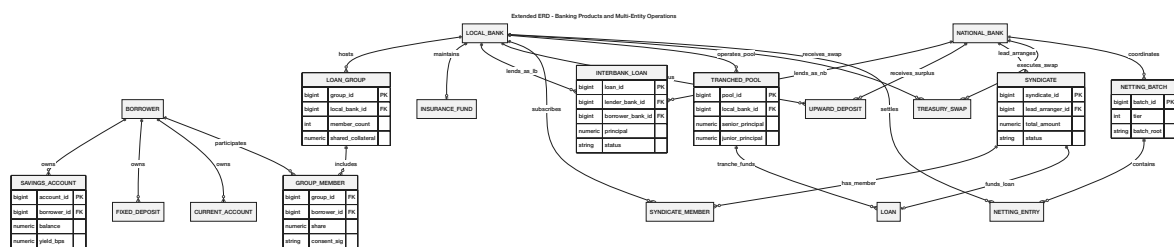
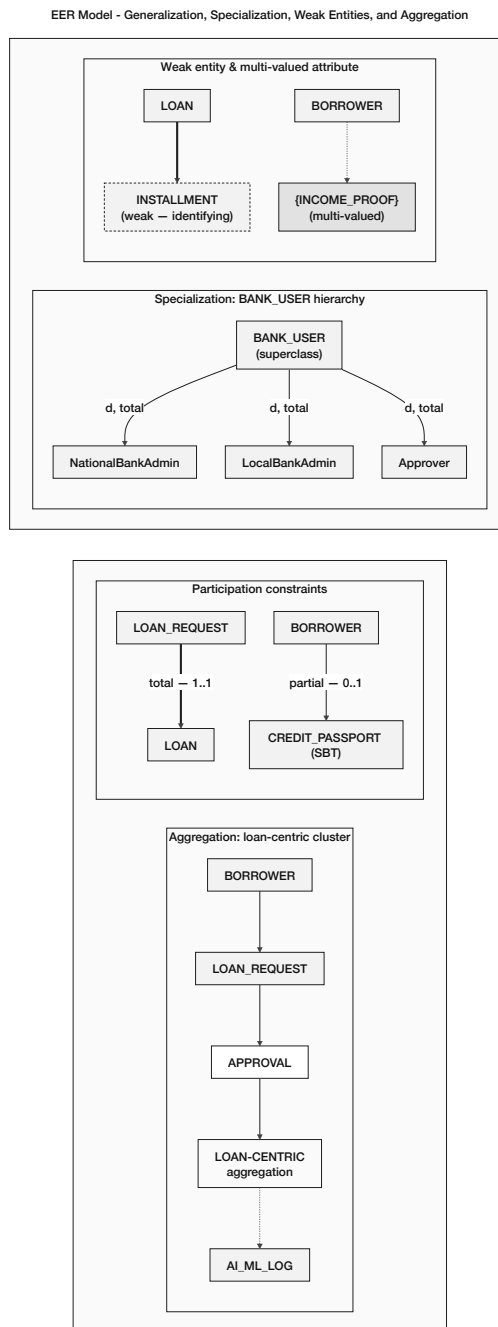
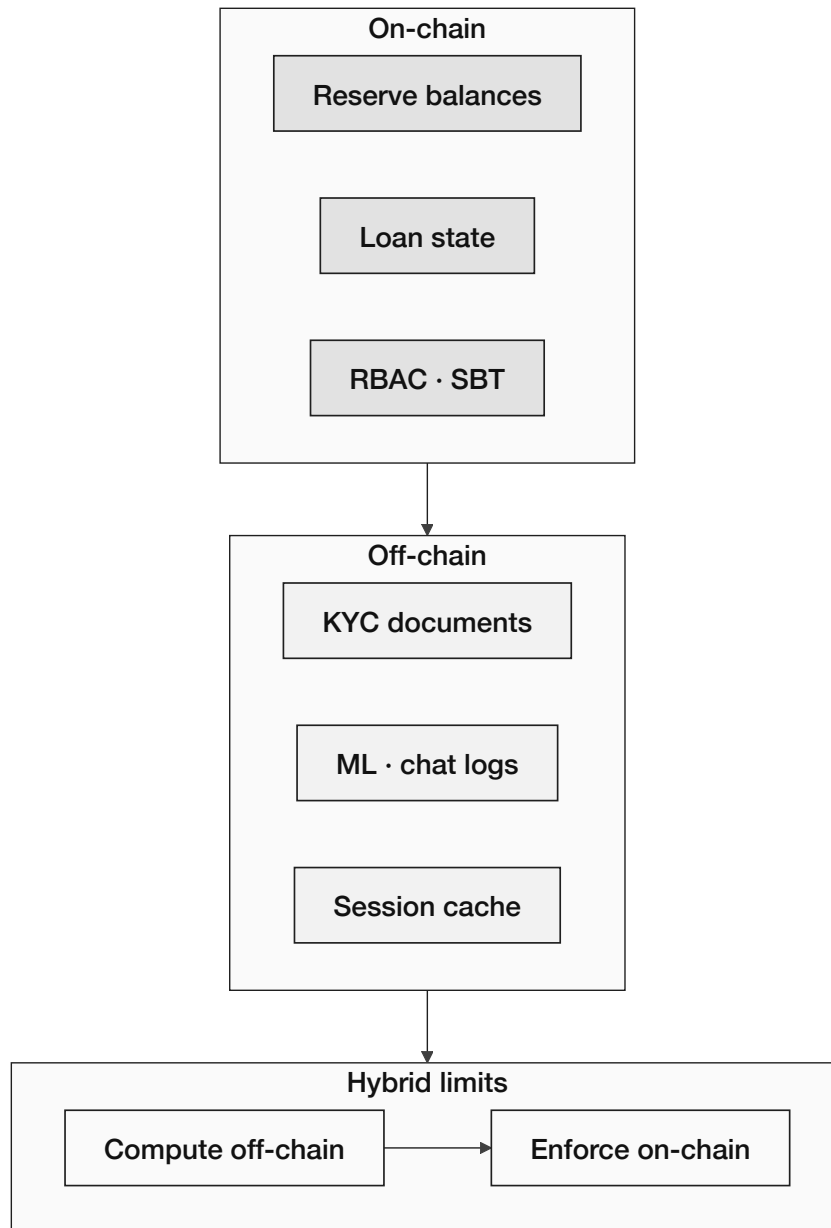


Figure 3.6: Extended ERD entities

**Figure 3.7:** Enhanced EER model

On-Chain vs Off-Chain Data Partitioning**Figure 3.8:** On-chain versus off-chain data partitioning

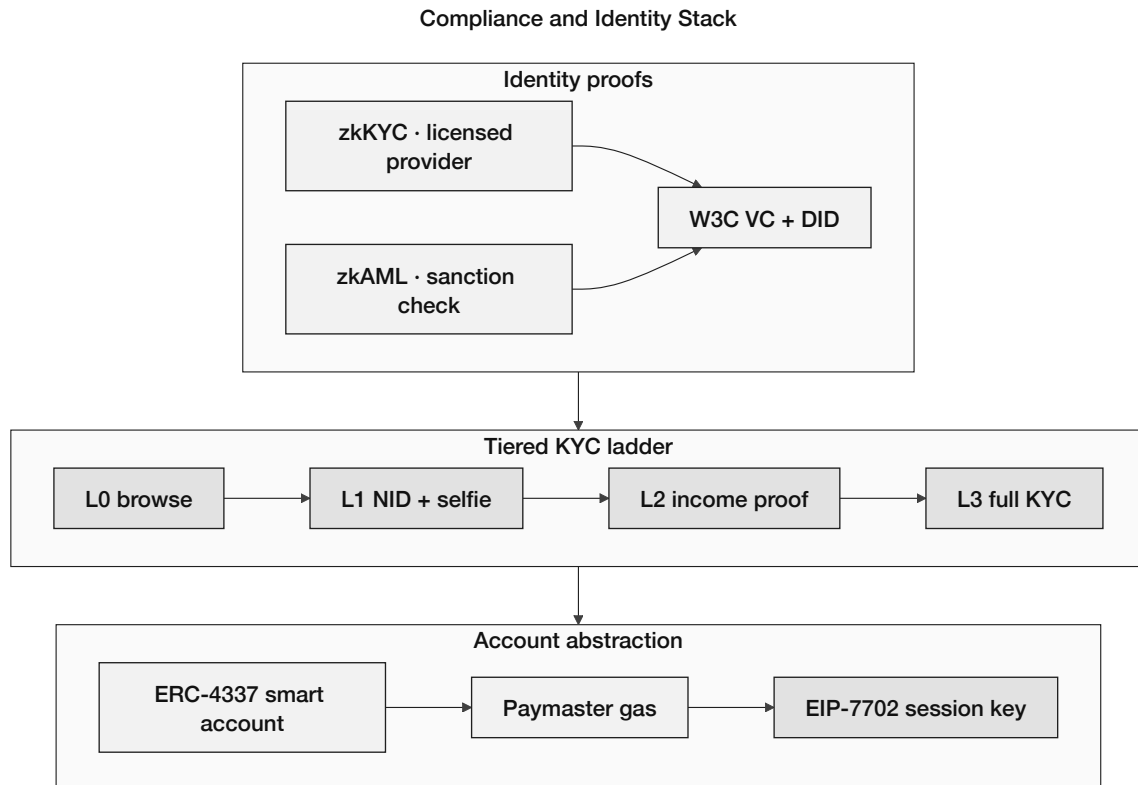


Figure 3.9: Compliance and identity stack

	Client (React UI / MCP Agent)	Local Bank (Tier 3)	National Bank (Tier 2)	World Bank Reserve (Tier 1)	AI Agent (Optional)
Action / Tool					
Submit loan request	✓	✓	○	○	---- ✓
Pay installment	✓	✓	○	○	---- ✓
Approve loan	○	✓	✓	○	---- ○
Approve account	○	✓	✓	○	---- ○
Freeze accounts	○	○	✓	✓	---- ○
View all rates	✓	✓	✓	✓	---- ✓
Read tools freely	✓	✓	✓	✓	---- ✓
Write tools (humans only)	○	✓	✓	✓	---- ○
✓ Allowed ○ Not allowed —— Full access (as per role) ---- Optional / limited (AI Agent)					

Figure 3.10: Actor permission matrix (primary actions)

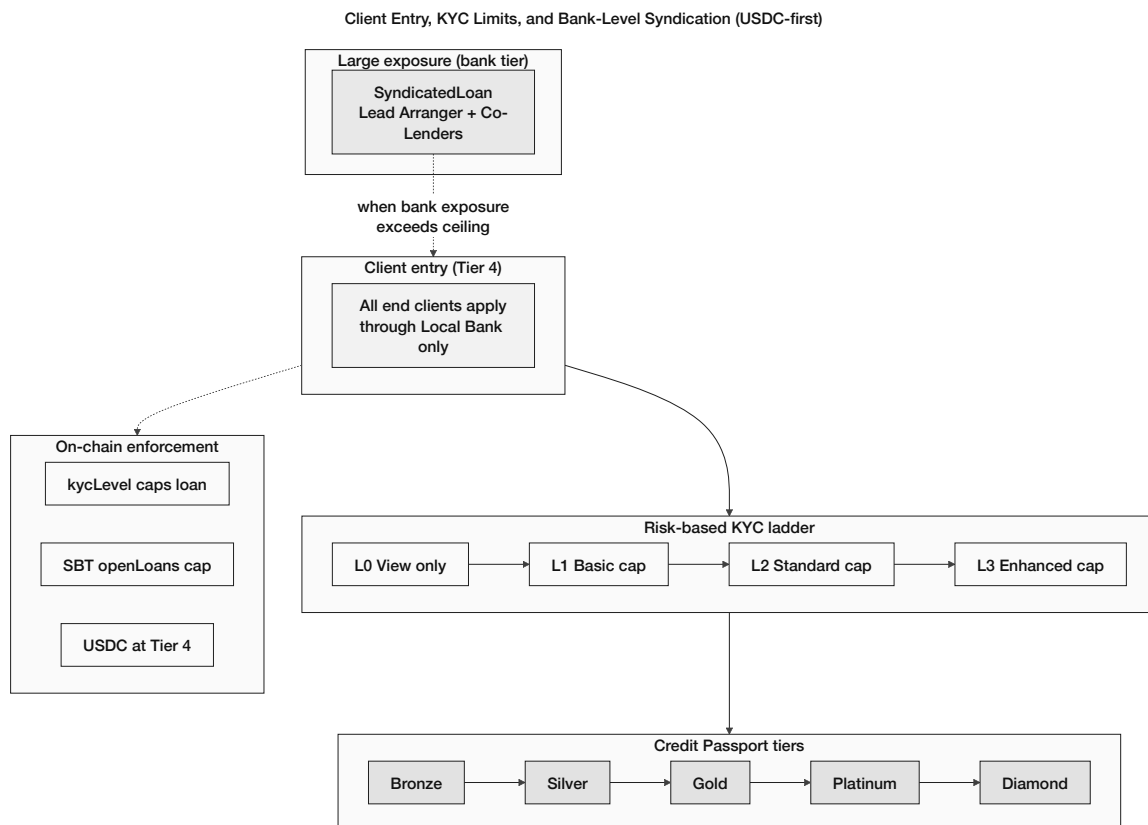


Figure 3.11: Client limits and bank-level syndication

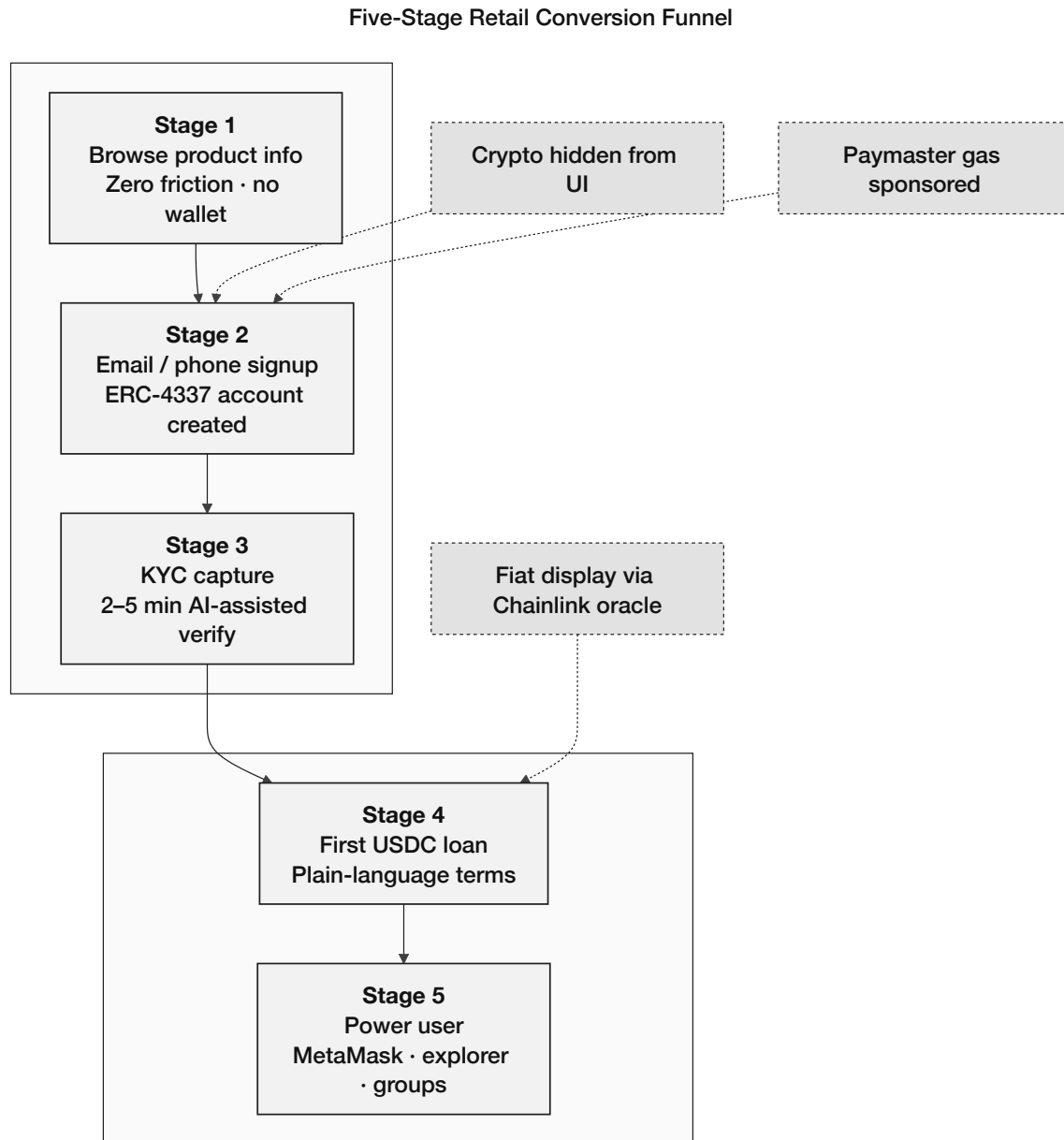
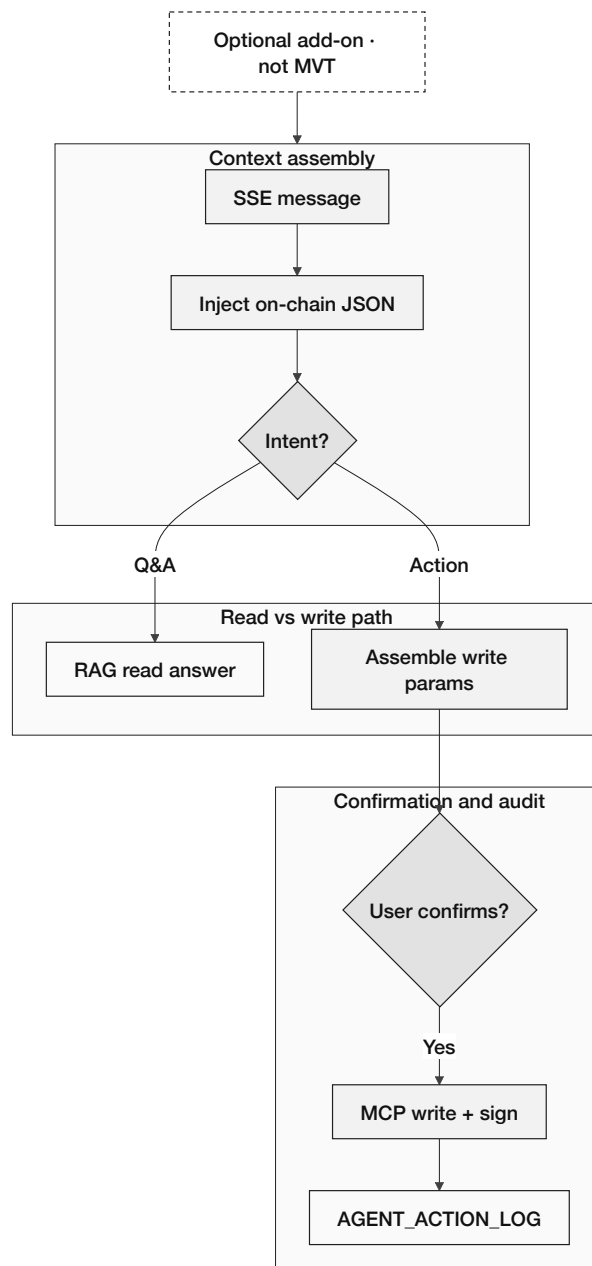


Figure 3.12: Five-stage retail onboarding funnel

Optional AI Agent Add-On — Six-Step Pipeline (Phase III–IV)

**Figure 3.13:** Optional AI agent add-on (Phase III–IV, excluded from MVT)

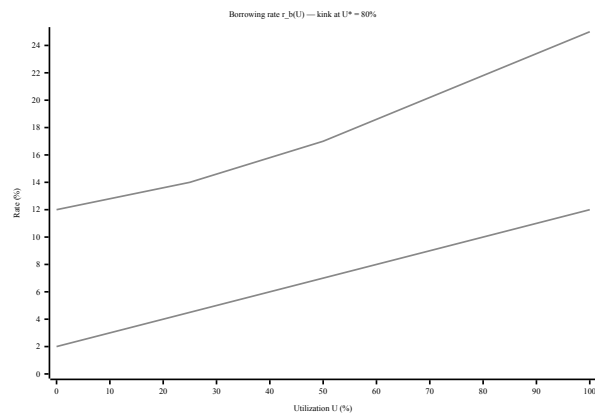


Figure 3.14: Kinked utilization-based borrowing rate

LiquidationEngine — Tier-Specific Health Factor and Lifecycle

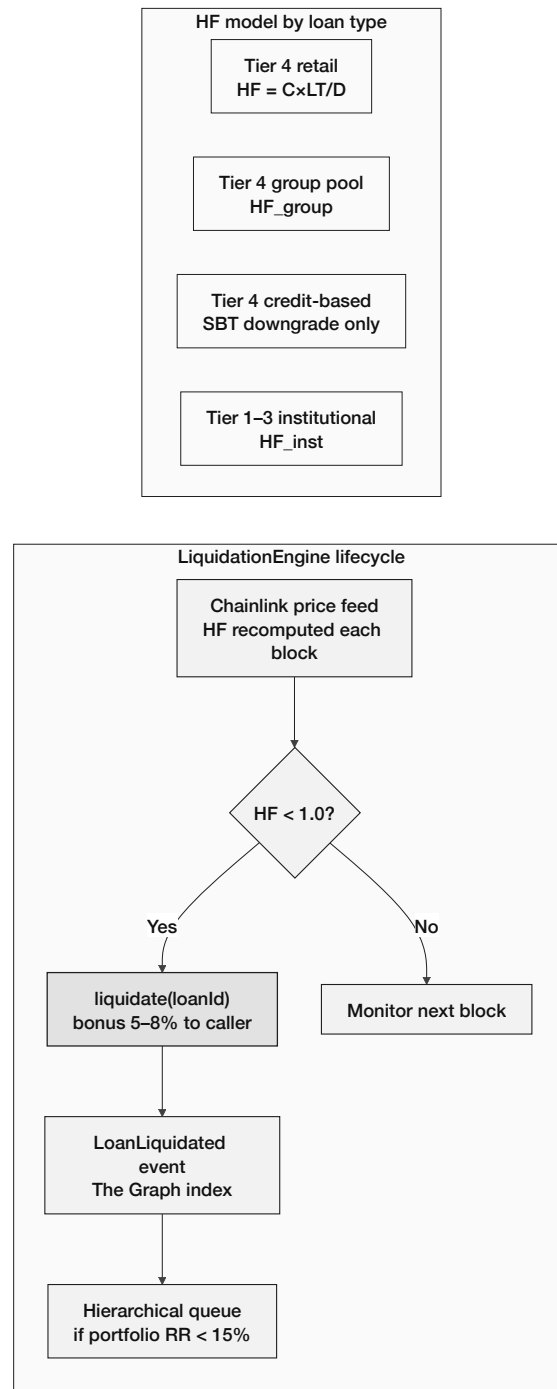


Figure 3.15: LiquidationEngine tier models and lifecycle

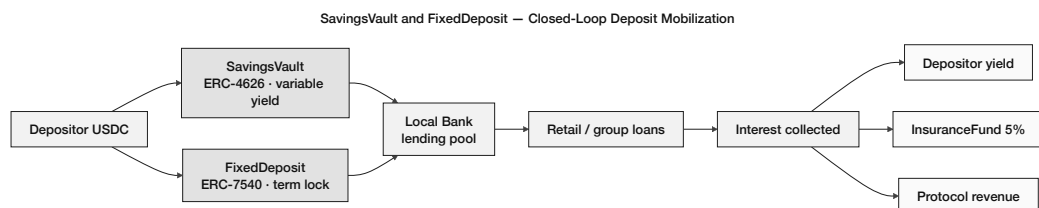
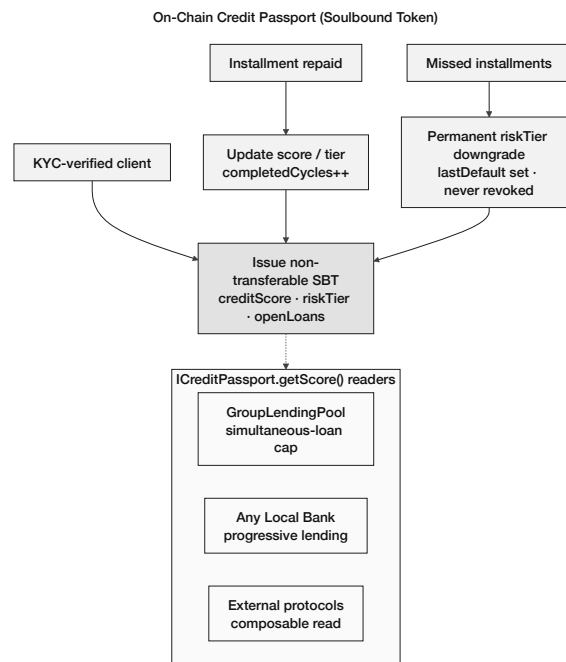


Figure 3.16: SavingsVault and FixedDeposit closed loop

**Figure 3.17:** On-chain Credit Passport SBT lifecycle

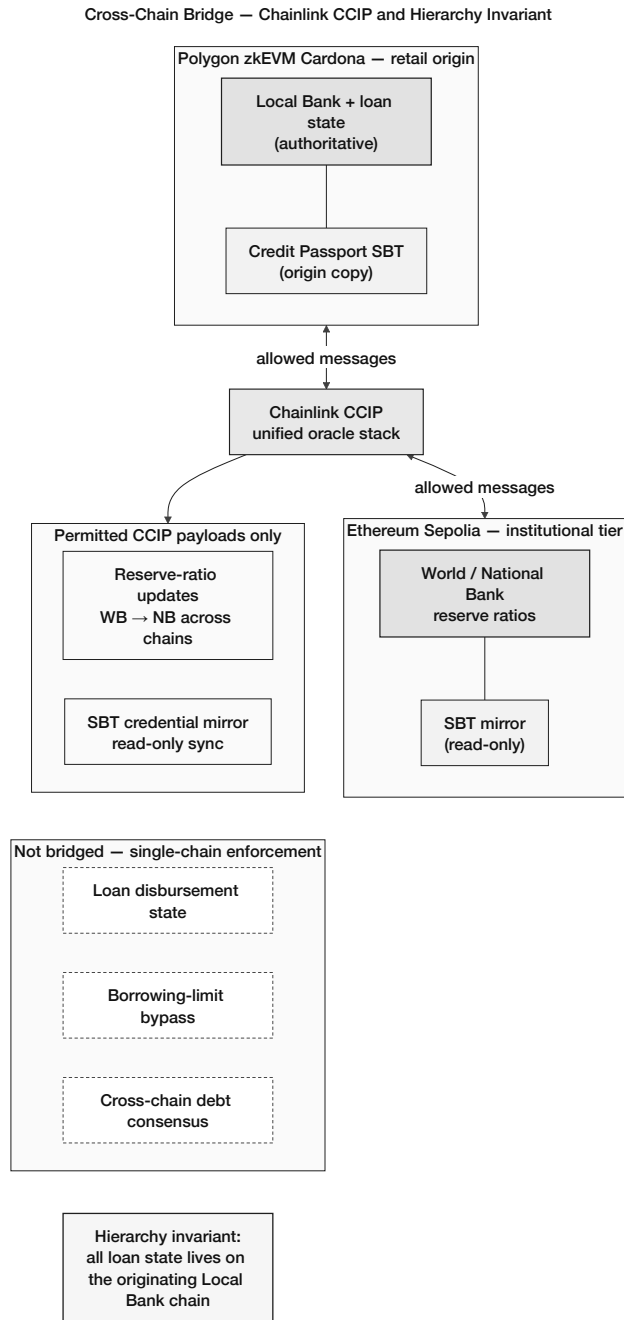


Figure 3.18: Chainlink CCIP cross-chain hierarchy invariant

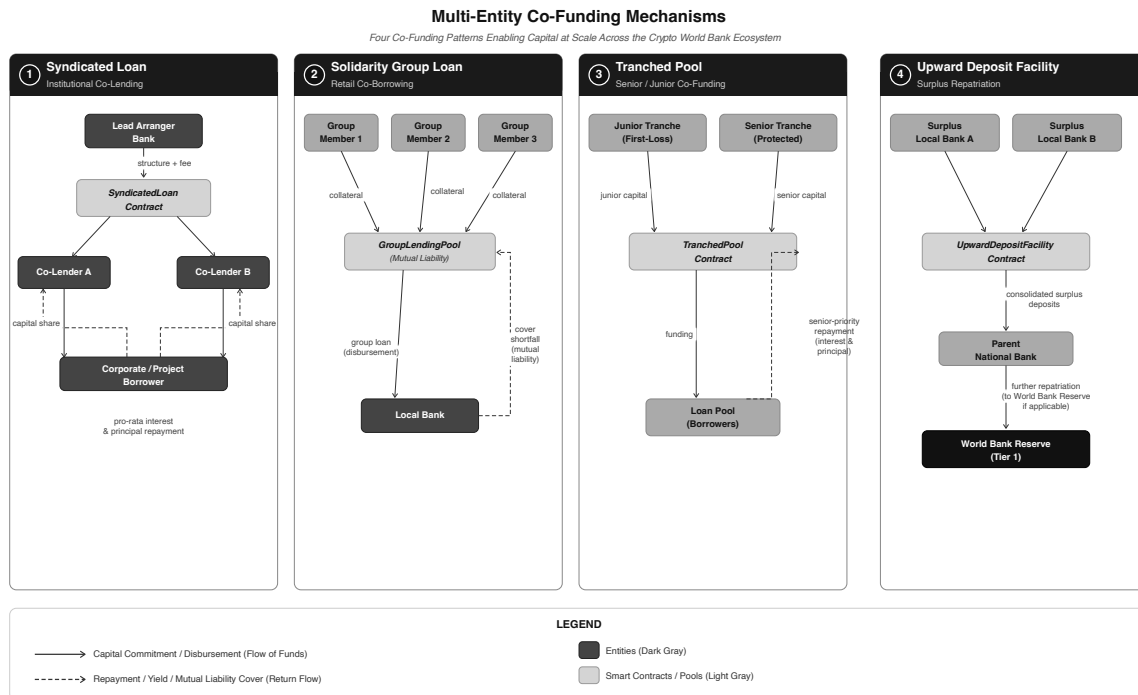


Figure 3.19: Multi-entity and cross-tier capital operations

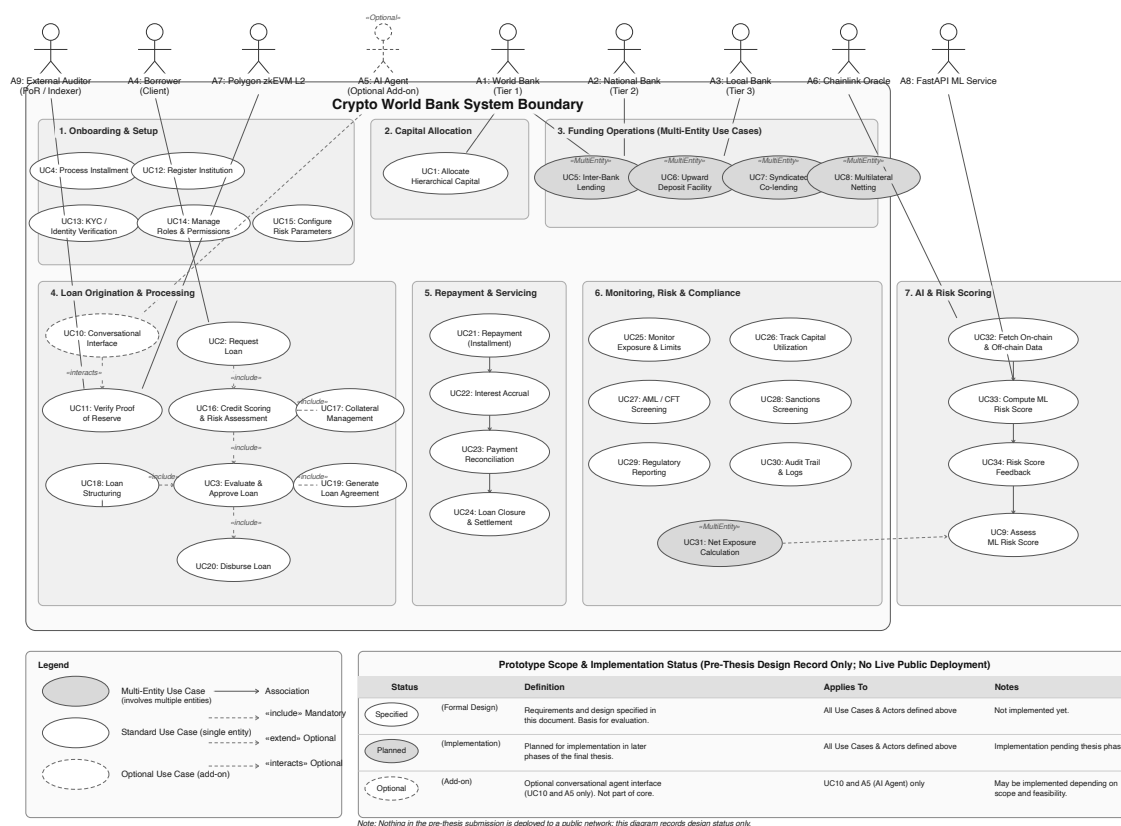


Figure 3.20: Use-case diagram (nine-actor taxonomy)

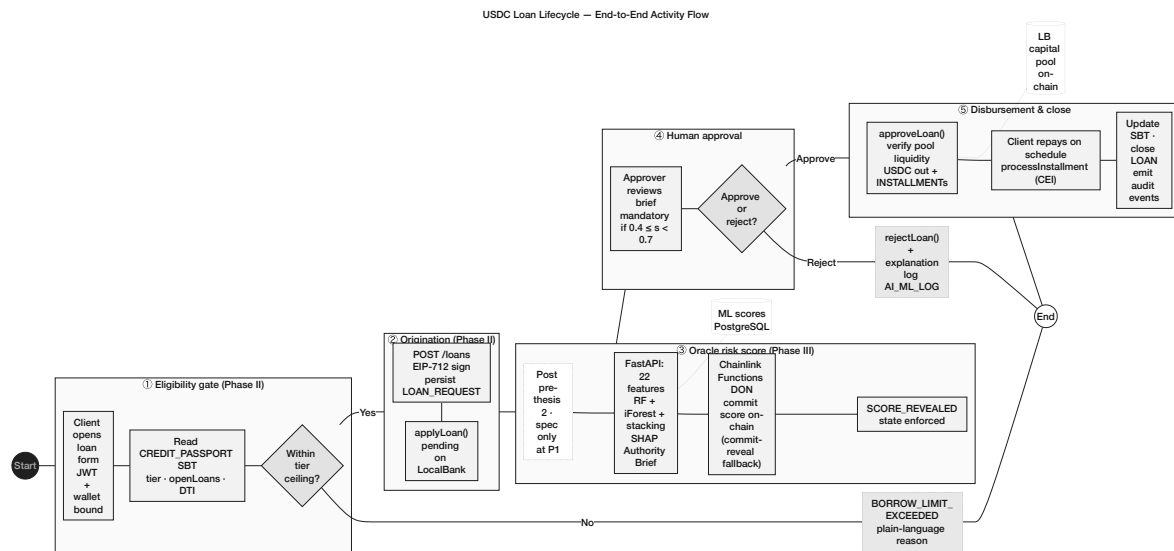


Figure 3.21: USDC loan lifecycle activity flow

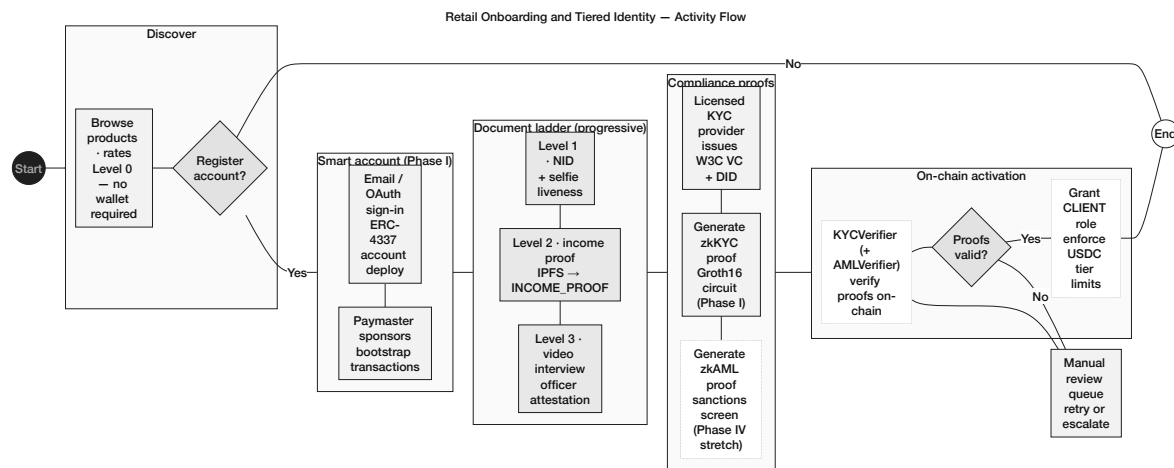


Figure 3.22: Retail onboarding and identity activity flow

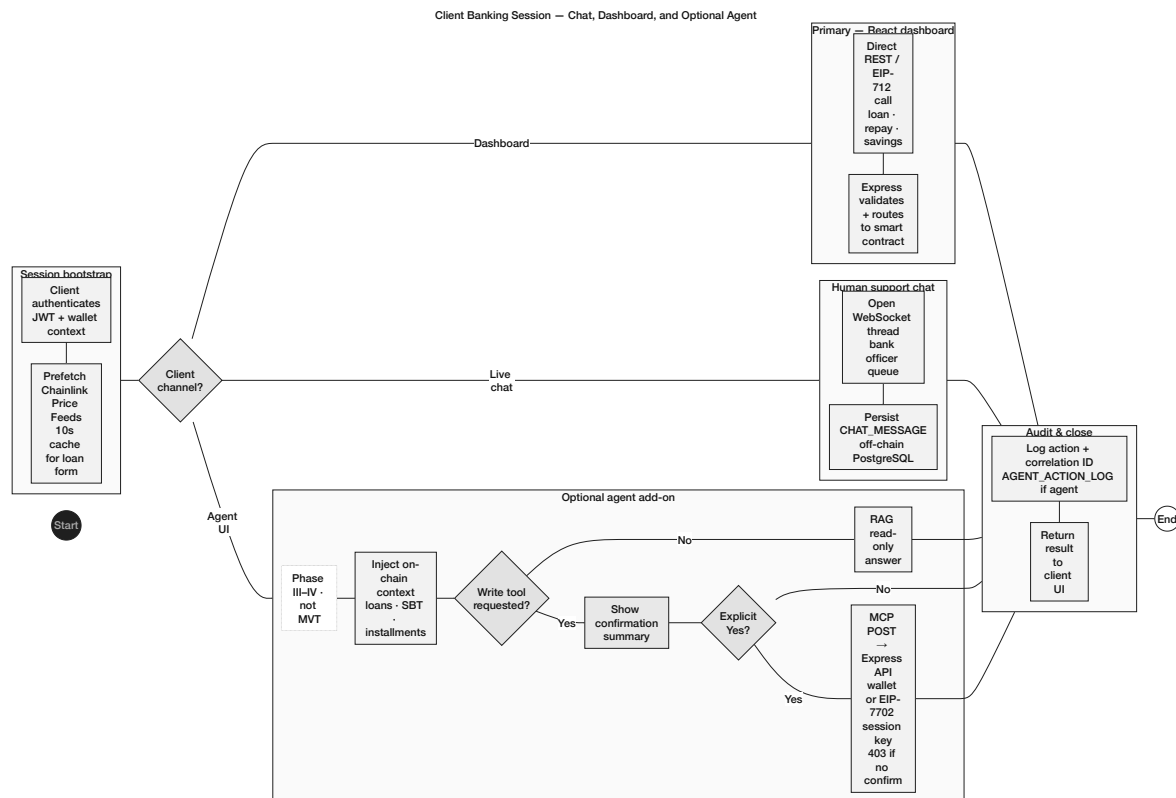


Figure 3.23: Client banking session activity flow

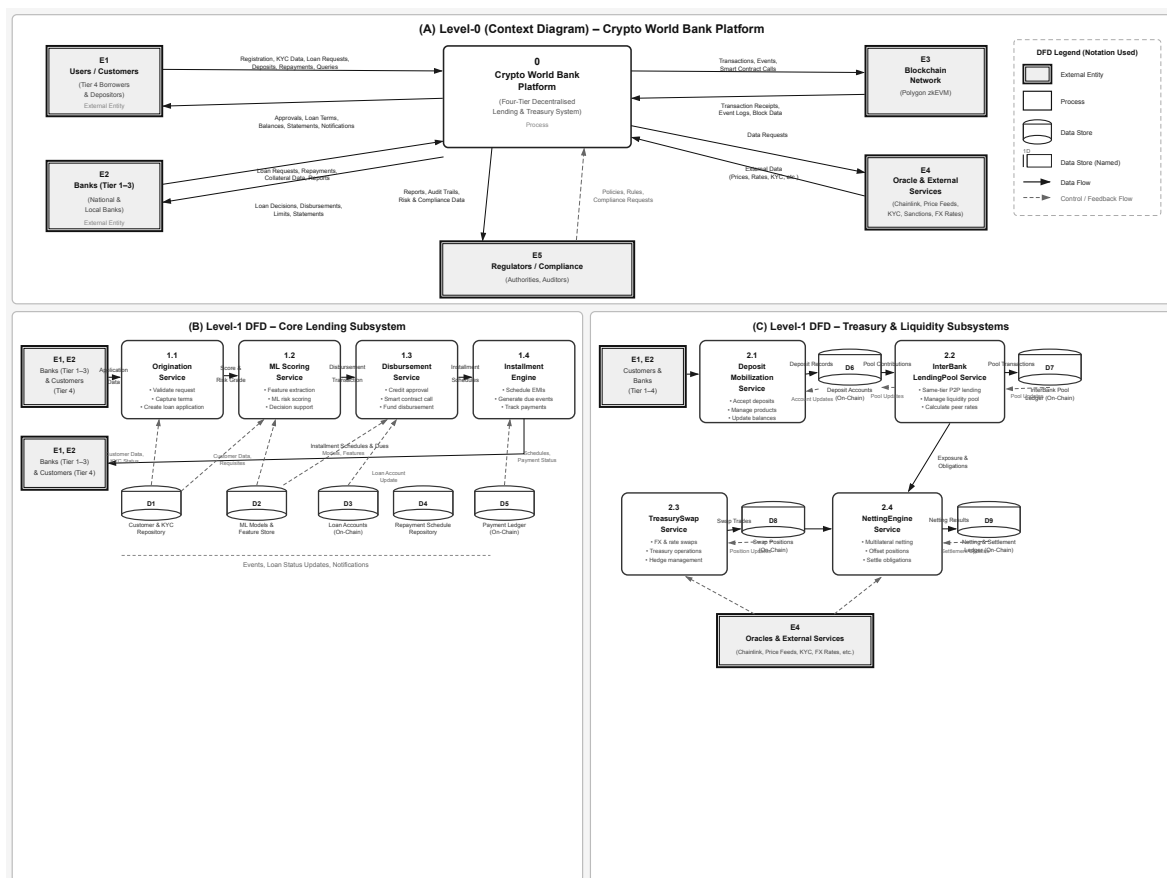


Figure 3.24: Data-flow diagrams

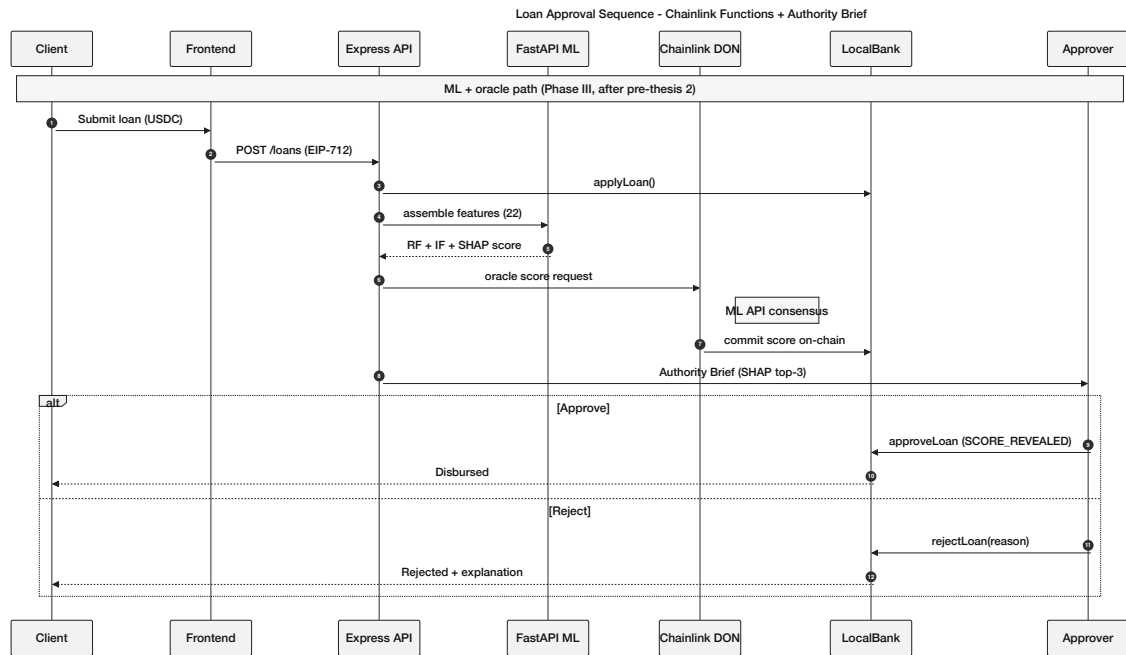


Figure 3.25: Loan approval sequence diagram

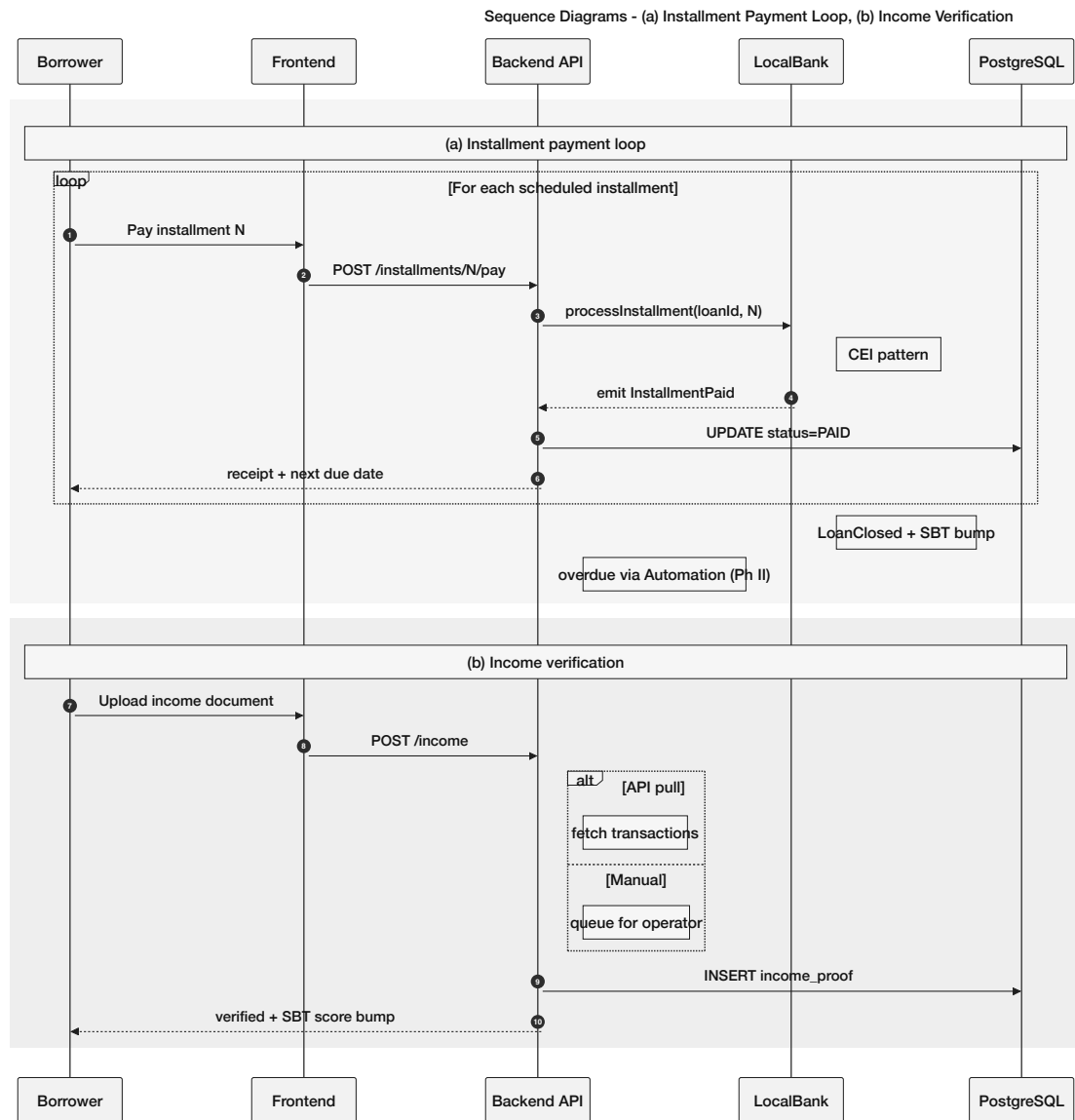


Figure 3.26: Sequence diagrams

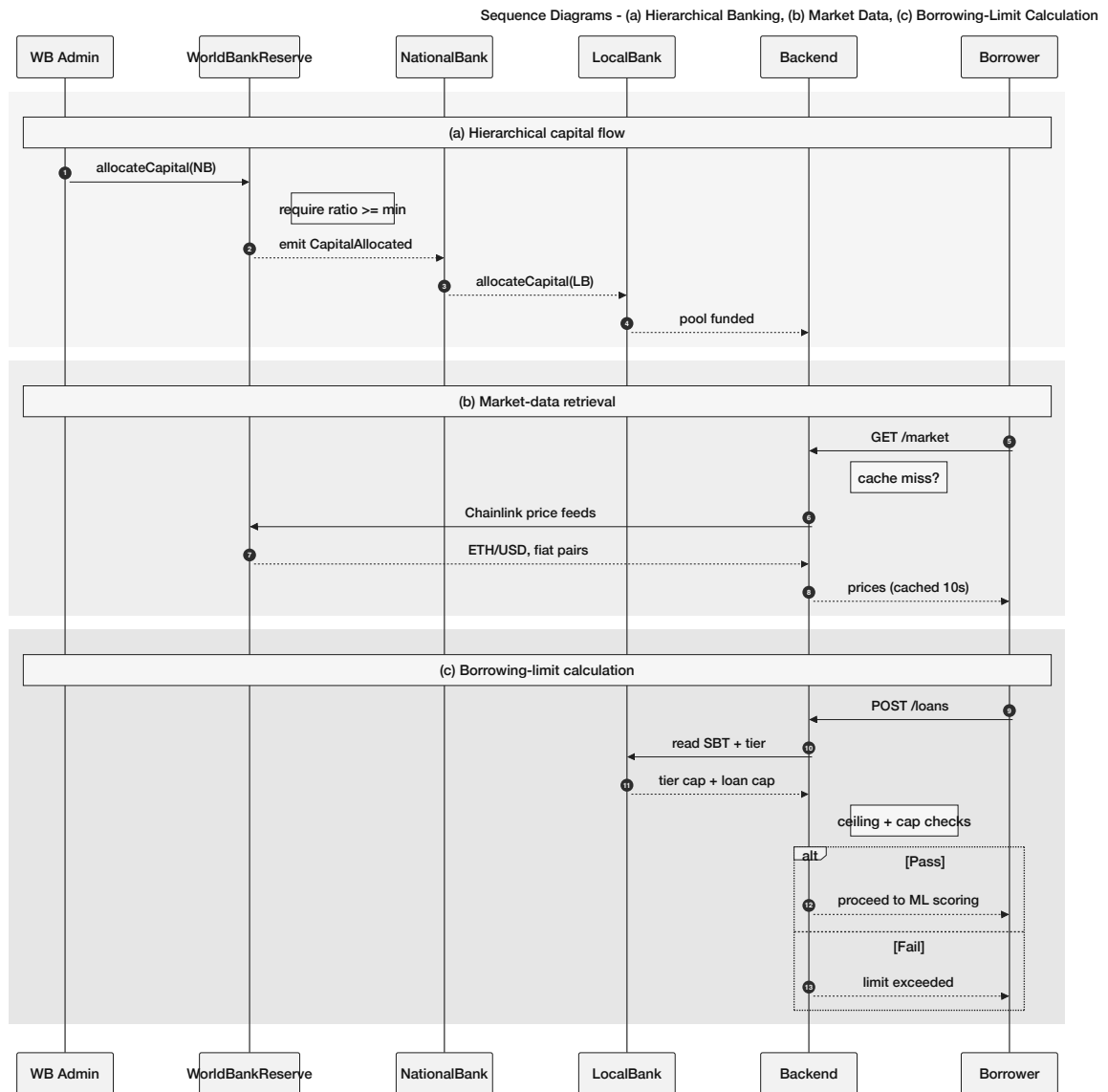
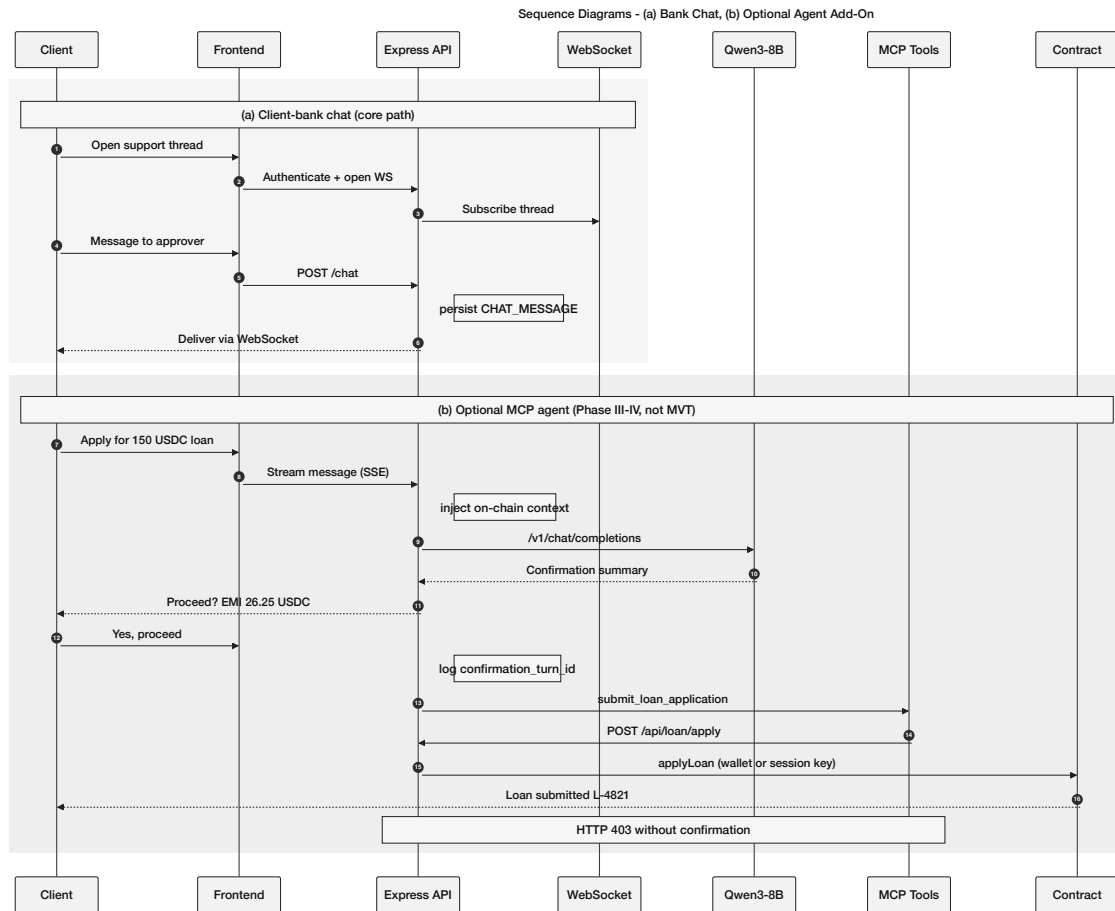


Figure 3.27: Sequence diagrams



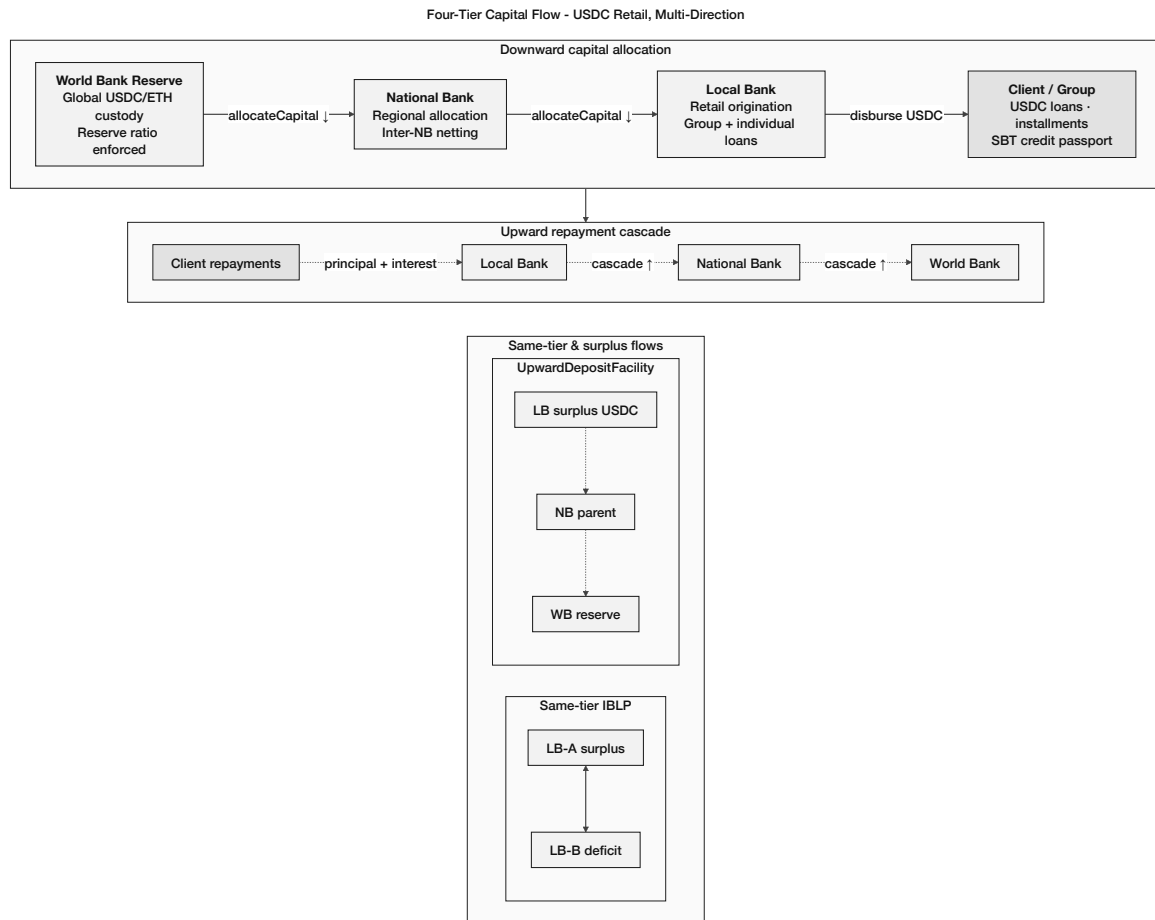


Figure 3.29: Four-tier hierarchical capital flow

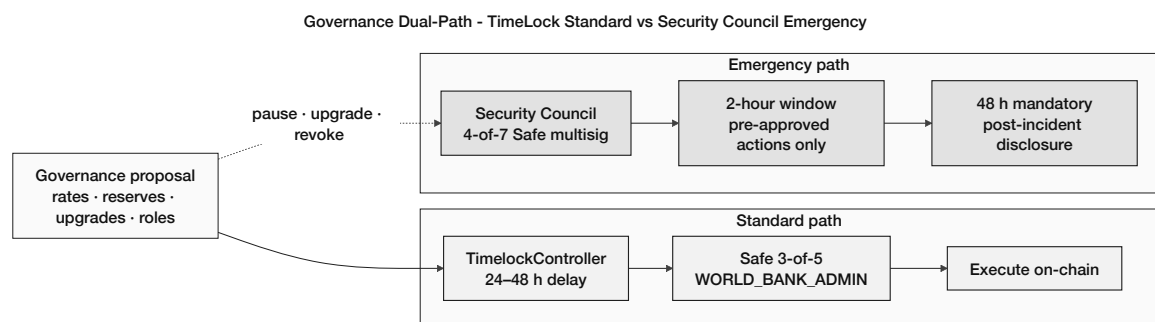


Figure 3.30: Governance dual-path

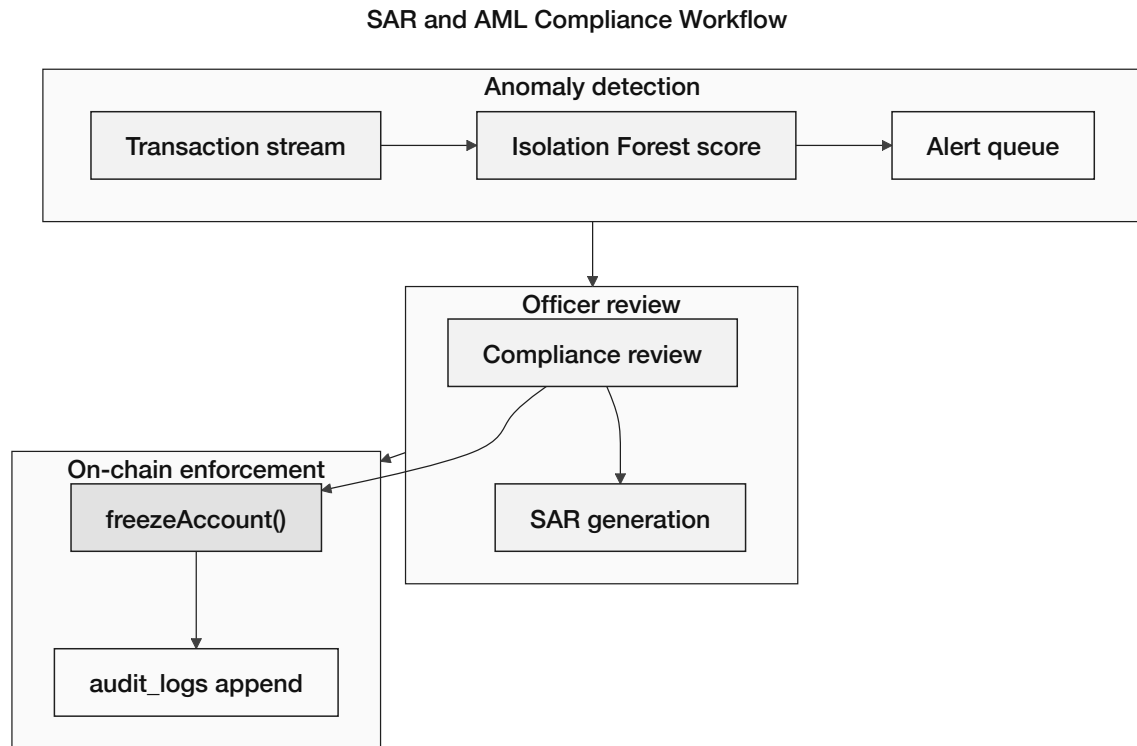
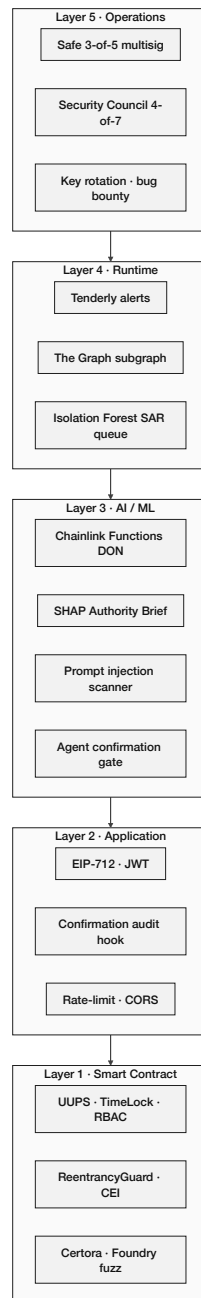


Figure 3.31: SAR and AML compliance workflow

Five-Layer Defense-in-Depth Security Architecture

**Figure 3.32:** Five-layer defense-in-depth security architecture

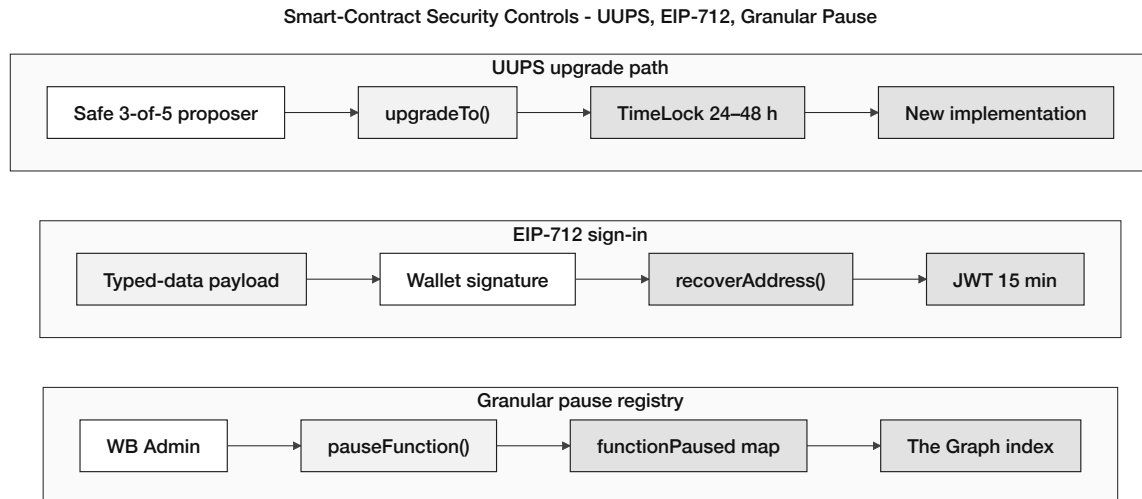


Figure 3.33: Smart-contract security controls

Chapter 4

Methodology

Lightweight **Agile/Scrum** across four implementation phases. **Core thesis** = on-chain banking + oracle ML; conversational agent is optional (Could-have).

4.1 MVT Checklist and Phase Plan

Graded pass line: (C1) four-tier contracts on testnet; (C2) GroupLendingPool spec + PoC; (C3) RF/IF/SHAP + Authority Brief after pre-thesis 2. Pre-thesis 1 delivers specification only—no trained models or live deploy.

Table 4.1: Four-phase implementation summary

Phase	Focus	Weeks	Key deliverables
I	Foundation	1–4	Contract hierarchy, DB schema, React shell
II	Core banking	5–9	Loan E2E, savings, group pool, automation
III	ML + oracle	10–13	Chainlink Functions, Authority Brief, dashboard
IV	Verification	14–16	Foundry sim, Certora stretch, testnet manifest

Table 4.2: Core vs. optional capabilities

Capability		Priority	Phase / interface
Four-tier smart contracts		Must (I–II)	React + wallet; Foundry tests
On-chain loan life-cycle		Must (II)	LoanController state machine
Chainlink tions oracle	Func-	Must (III)	FastAPI → DON → on-chain score
RF + IF + SHAP		Must (III)	Authority Brief; audit log
MCP conversational agent		Could (III–IV)	Alternate API path only

4.2 Design Decisions

Table 4.3: Design decisions (selected)

Area	Choice	Alternative	Criterion
Network	Polygon zkEVM Cardona	Amoy PoS	ZK validity proofs, low gas
Contracts	Solidity EVM	Solana	Tooling, testnet cost
Fraud ML	Random Forest + IF	XGBoost	SHAP compatibility
Oracle	Chainlink Functions	Commit-reveal only	Decentralised trust
Database	PostgreSQL	SQLite	3NF, async API

4.3 Methodology Diagrams (Full Set)

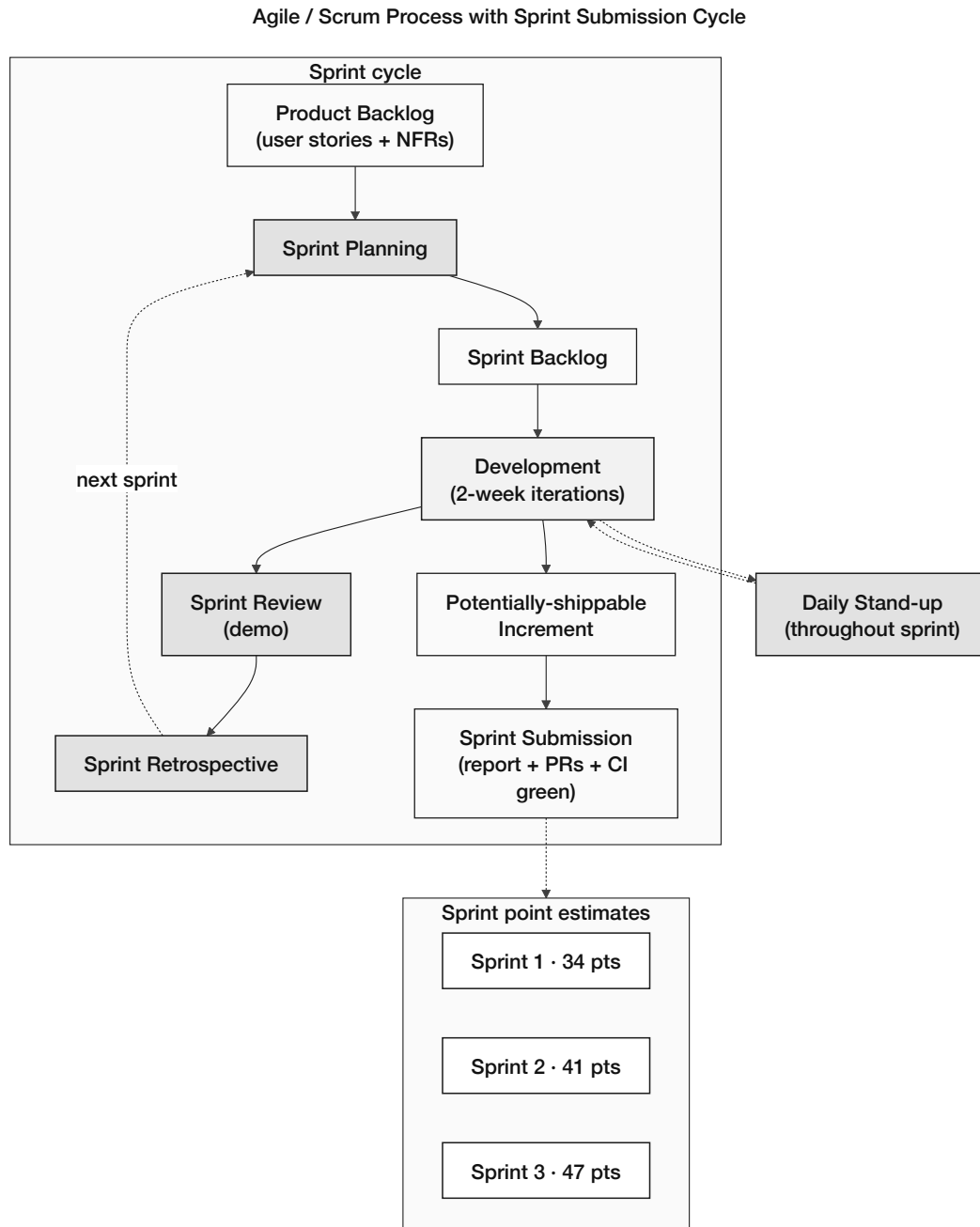
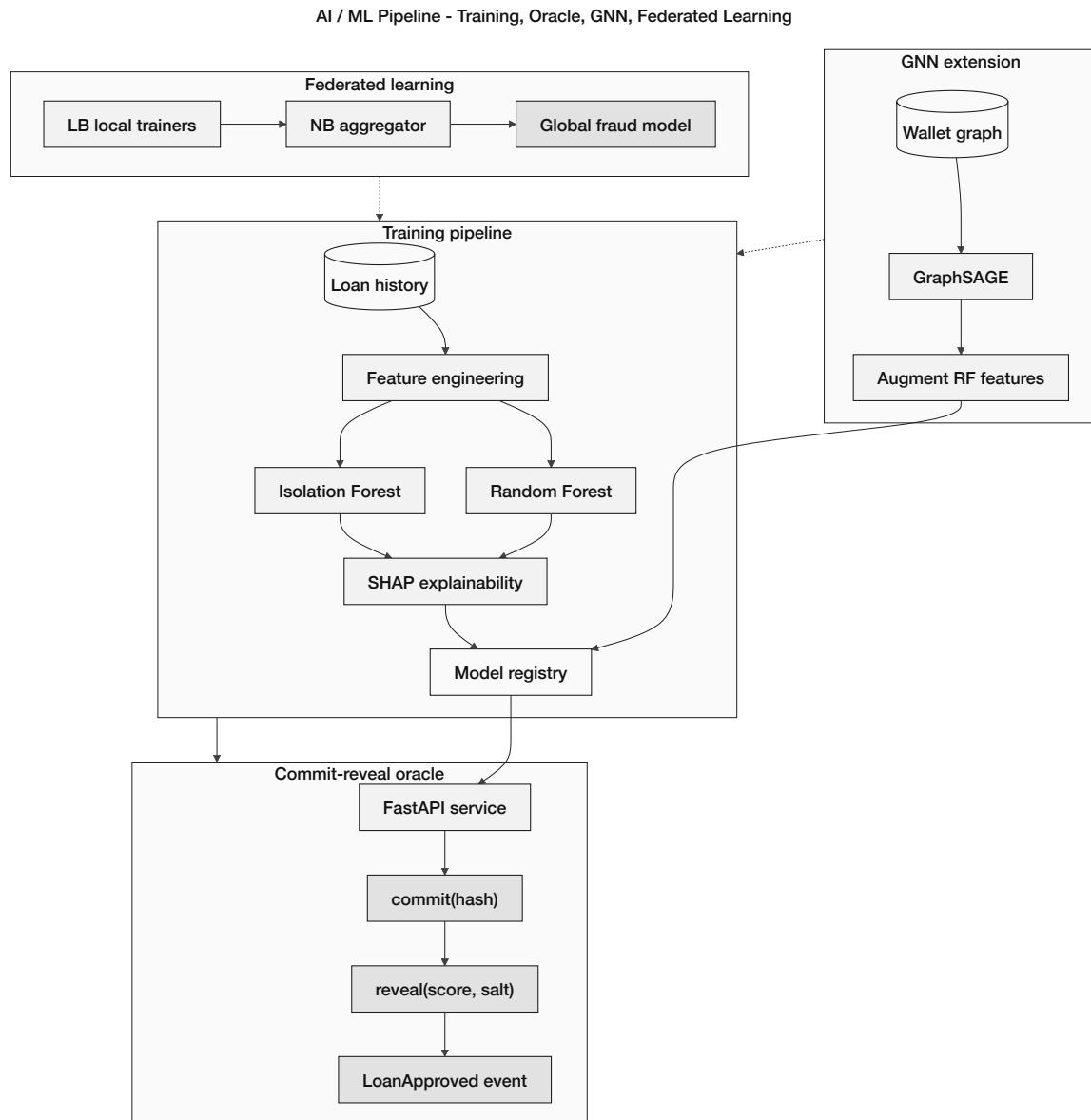


Figure 4.1: Agile / Scrum development process

**Figure 4.2:** AI/ML pipeline wiring

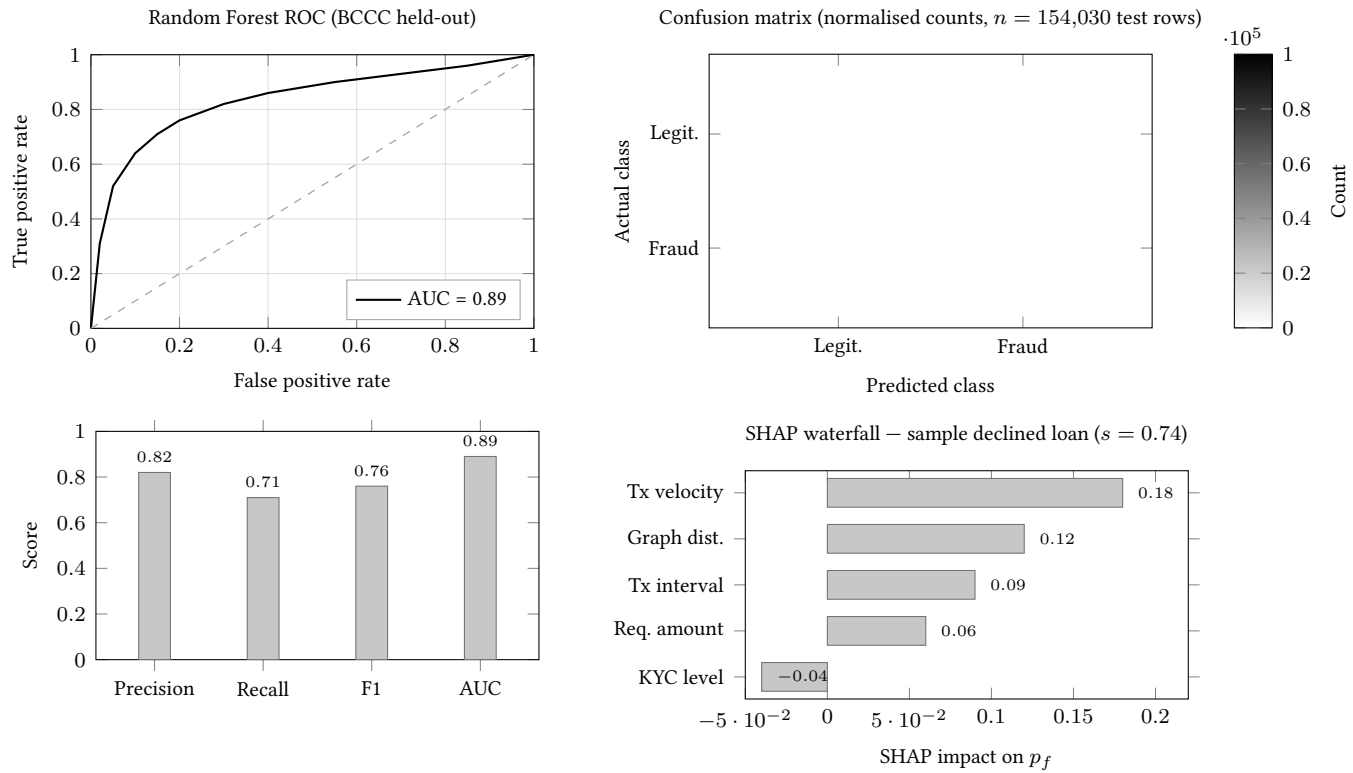


Figure 4.3: Planned ML evaluation layout (post pre-thesis 2)

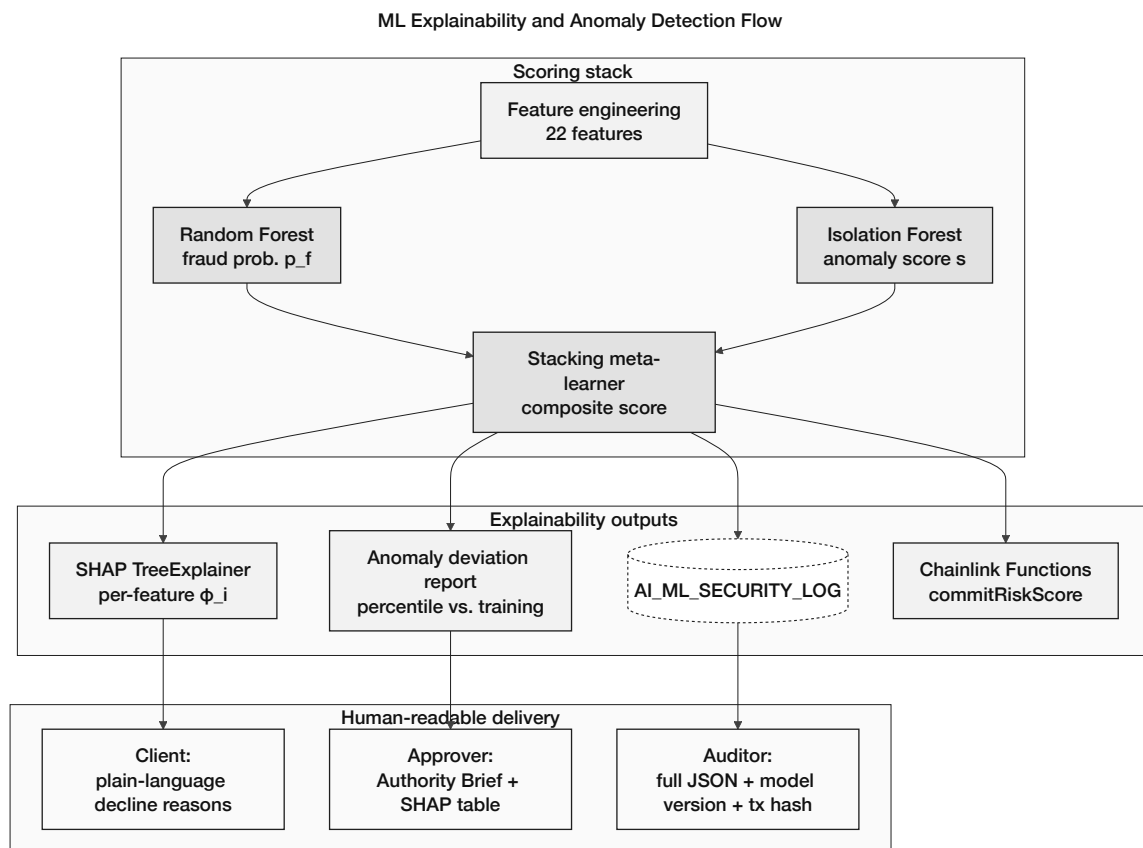


Figure 4.4: Explainability and anomaly-detection flow

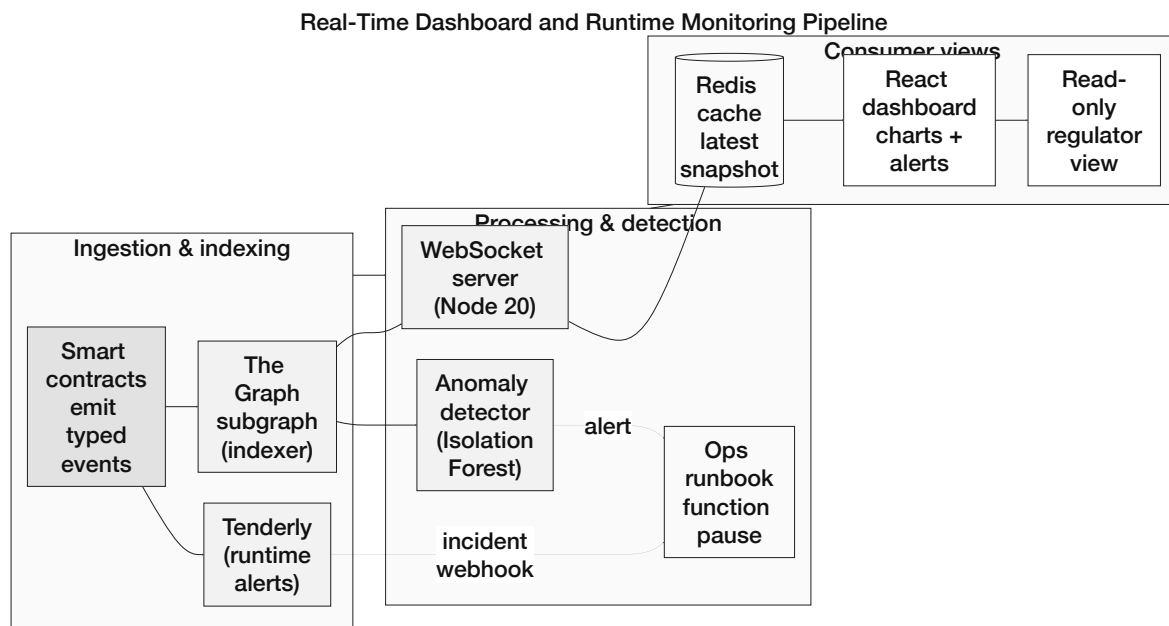


Figure 4.5: Real-time dashboard and runtime monitoring pipeline

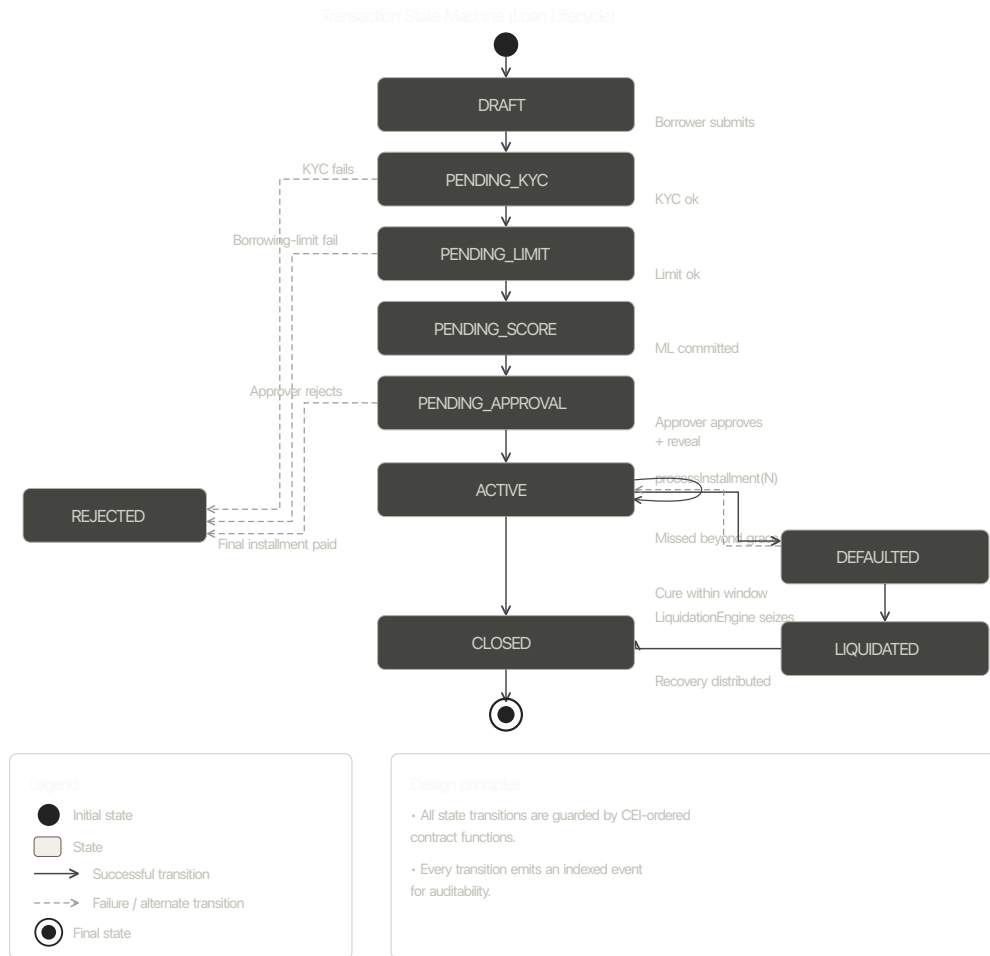


Figure 4.6: DRAFT → PENDING_KYC → PENDING_LIMIT → PENDING_SCORE → PENDING_APPROVAL → ACTIVE → {CLOSED, DEFAULTED} → {LIQUIDATED, cured-back-to-ACTIVE}, with every transition guarded by a CEI-ordered contract function and every transition emitting an indexed event.

Figure 4.6: Transaction state machine of the loan lifecycle

Figure 4.8: Four-phase Implementation Roadmap with SDLC Stage Mapping

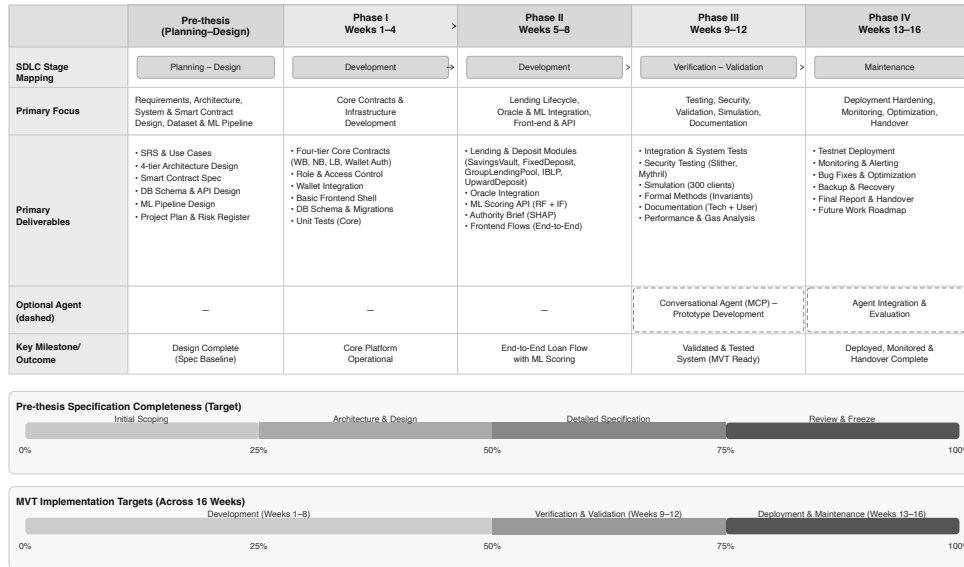


Figure 4.7: Four-phase roadmap with SDLC mapping and progress

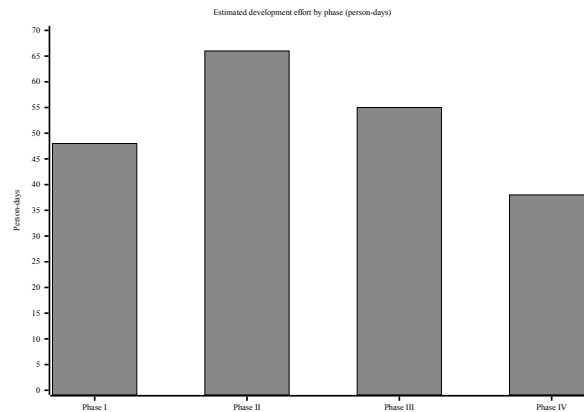


Figure 4.8: Development effort by implementation phase

Optional Conversational Agent Add-On (compact)

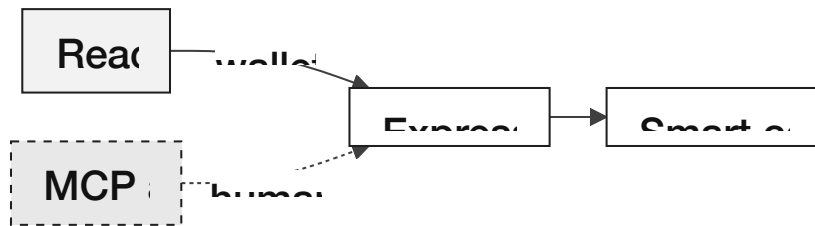


Figure 4.9: Optional conversational agent add-on

Planned Development and Verification Toolchain

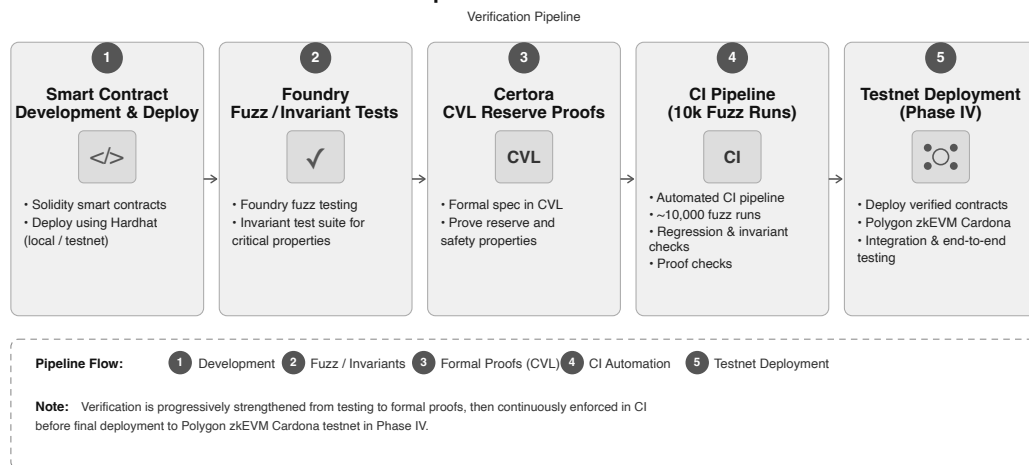


Figure 4.10: Planned development and verification toolchain

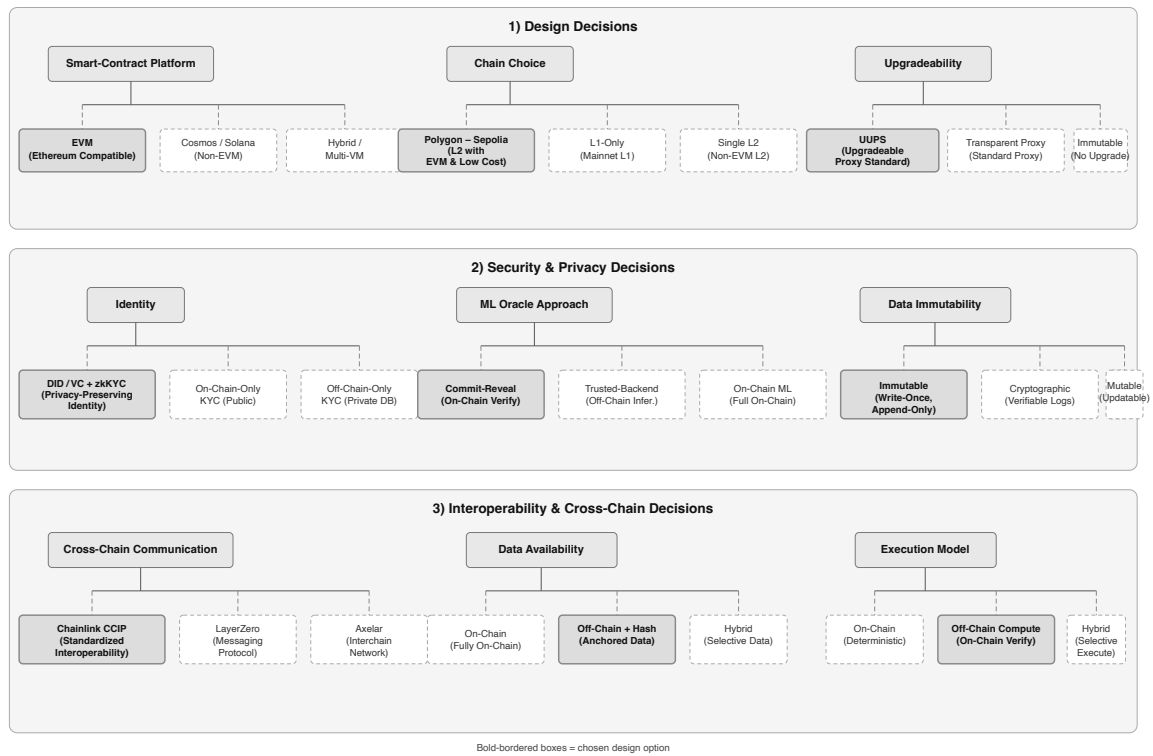


Figure 4.11: Key design decisions and alternatives considered

Chapter 5

Evaluation and Discussion

Authoring instructions — Chapter 5 (short version)

Goal: Show feasibility “slightly” for pre-thesis 1; label all numbers as *planning projections* until pre-thesis 2 evidence exists.

Must include (in order):

1. **Performance evaluation** (1 short paragraph): pre-thesis 1 = specs only; live gas/ML benchmarks deferred to pre-thesis 2; point to MVT checklist.
2. **Analysis of design solutions** (1 paragraph): justify stack (zkEVM L2, Solidity, PostgreSQL, Chainlink Functions); optional agent outside MVT.
3. **Final design adjustment** (1 paragraph): four locks — stablecoin-first, L2 gas, ML as decision-support only, MVT scope boundary.
4. **Feasibility analysis** — keep all **four tables**: technical, economic (\$0 prototype), risk taxonomy, regulatory/MiCA mapping.
5. **Statistical analysis** — keep revenue-assumption table, ETH-sensitivity table, default-scenario table, and both APR/revenue figures; open with “planning projections, not observed data.”
6. **Comparisons and relationships** — keep market-sizing, customer-segment, and partner-role tables; **do not** duplicate the Ch. 2 protocol-comparison table.
7. **Discussion** (1 short section): five explicit limitations (compound novelty, BCCC baseline, testnet scope, group lending $\neq$ BRAC, oracle drift).

Cut from full Ch. 5: long prose repeats, duplicate competitor table, appendix-level detail. Target: $\approx$ 8–12 pages with tables + 2 figures.

5.1 Performance Evaluation

Pre-thesis 1 delivers specifications and planned checks, not live benchmarks. Component and economic feasibility are in Section 5.4; RQ-level evaluation follows Section 5.1 after pre-thesis 2 (testnet deployment, gas measurement, ML metrics, MVT demo). On L2, projected gas per loan stays below interest earned; default stress is in Table 5.7.

5.2 Analysis of Design Solutions

Stack choices were scored on ecosystem maturity, zero-cost prototype constraints, and auditability (Table 4.3, Section 4.2). The selected path pairs EVM Solidity on zkEVM L2, PostgreSQL + FastAPI off-chain, and Chainlink Functions (with commit-reveal as an interim relay) for ML scores. The optional MCP agent and EIP-7702 session keys remain Phase III–IV extensions outside the MVT.

5.3 Final Design Adjustment

The final pre-thesis design locks four adjustments before implementation: (i) **stablecoin-first** retail lending (USDC/USDT, Section 1.8) to avoid ETH liability volatility; (ii) **L2 deployment** for micro-loan gas economics; (iii) **ML as decision support only**—human approvers and on-chain gates, no auto-disbursement; (iv) **MVT scope** limited to the three-tier lending core, database schema, and dashboard—syndicated/tranched modules, full regulatory filing, and the conversational agent are deferred (Table ??).

5.4 Feasibility Analysis

5.4.1 Technical Feasibility

Each layer is specified against mature tooling; nothing requires bespoke infrastructure beyond standard testnets and open-source stacks (Table 5.1).

Table 5.1: Technical feasibility assessment

Component	Assessment	Evidence
Smart contracts	Feasible (specified)	Solidity; Hardhat; OpenZeppelin; Phases I–II
Frontend DApp	Feasible (specified)	React 18 + Wagmi/RainbowKit
Blockchain deployment	Feasible (planned)	Polygon zkEVM Cardona + Sepolia
AI/ML integration	Feasible with constraints	RF + IF + SHAP; Chainlink Functions Phase III
Optional agent	Feasible (deferred)	MCP + Qwen3-8B; Phase III–IV; outside MVT
Database backend	Feasible	PostgreSQL 20-entity schema (3NF); FastAPI

5.4.2 Economic Feasibility

The academic prototype can run at **zero direct cost**; scaled hosting is a separate pilot concern (Table 5.2). L2 gas keeps small loans viable versus mainnet; spread revenue and default scenarios are modelled in Section 5.5.

Table 5.2: Economic feasibility — zero-cost prototype

Cost Category		Estimate	Notes
Blockchain	deployment	\$0	Public testnets
Frontend	/ backend	\$0	Vercel / Render free tiers or localhost hosting
AI/ML training		\$0	Local machine or Colab free tier
Development tools		\$0	Hardhat, VS Code, Git (open-source)
Total prototype		\$0	Pilot ops ≈\$500–\$2,000/month at scale

5.4.3 Risk Feasibility

Main delivery risks and mitigations are summarised in Table 5.3; smart-contract and regulatory items dominate severity.

Table 5.3: Risk taxonomy and mitigation

Risk Category		Description	Severity	Mitigation
Partner	non-cooperation	Key partners decline to join	Medium	Testnet pilots; open-source replication
Smart	contract	Logic exploit	High	OpenZeppelin; audits; pause mechanism
Regulatory	ad- versity	Jurisdiction blocks re-tail crypto	Medium	Testnet-only prototype; sandbox path
AI/ML	model	Fraud model drift	Low	Retraining; human-in-the-loop; SHAP review

5.4.4 Regulatory Feasibility

Retail stablecoin lending must align with MiCA, GENIUS Act, and EU AI Act expectations; CWB uses authorised stablecoins and keeps PII off-chain (Table 5.4). Full conformity files and national licensing remain future work; pre-thesis scope is testnet demonstration only.

Table 5.4: Regulatory design controls (selected)

Statute / Article	Requirement	CWB Control
MiCA Art. 16	Authorised EMT issuer	Uses USDC/USDT; does not issue
MiCA Art. 36	1:1 reserve backing	Issuer attestations before pool acceptance
EU AI Act (credit scoring)	High-risk AI from Aug. 2026	Human approver gate; SHAP + audit log
GDPR Art. 17	Right to erasure	PII off-chain only; hashes on-chain

5.5 Statistical Analysis

The tables below model spread revenue, ETH-price sensitivity, and default stress at full-deployment scale (planning assumptions only; not observed platform data). Interest tiers follow the 3%/5%/8% hierarchy (Figure 5.2).

Table 5.5: Revenue projection assumptions

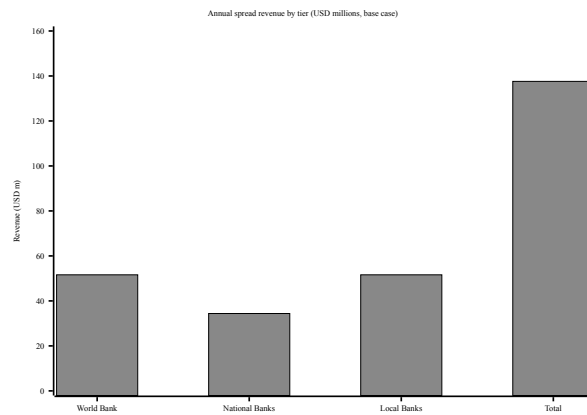
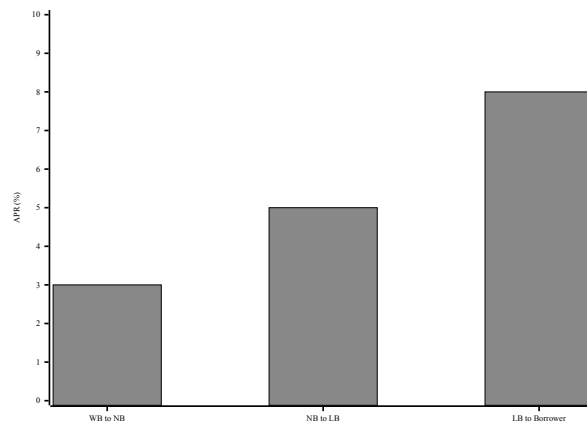
Parameter	Value
Reference ETH price	\$2,500 (planning mid-point, Feb 2026)
World Bank → National Bank	3% APR
National Bank → Local Bank	5% APR
Local Bank → Client	8% APR
Average loan term	12 months
Default rate provision (base)	8%
Origination fee	0.25% per disbursement

Table 5.6: ETH-price sensitivity of spread revenue

ETH Price	Loan Base	Spread Revenue	Incl. Fees (est.)
\$2,000	\$1.376B	\$110.1M	\$120–128M
\$2,500 (base)	\$1.720B	\$137.6M	\$150–159M
\$3,000	\$2.064B	\$165.1M	\$180–191M

Table 5.7: Default-rate sensitivity

Scenario	Default rate	Annual loss	Basis
Optimistic	3.7%	\$7.4M	Mature MFI benchmarks
Base case	8%	\$16M	New platform, limited history
Stress	15%	\$30M	Thin credit history, volatile collateral

**Figure 5.1:** Annual revenue projection (10-year maturity)**Figure 5.2:** Hierarchical interest-rate spread (APR)

5.6 Comparisons and Relationships

Table 5.8: Market segments with supporting data

Segment	Description	Estimated Scale
TAM	Global DeFi lending; remittances; MSME gap [8, 15, 20]	\$55B – 5T+
SAM	Institutional hierarchical lending; emerging-market credit	\$5B – 15B
SOM	Testnet pilots; sandbox partnerships	\$50 – 200M

Table 5.9: Target customer segment profile

Characteristic	Description
Primary users	Tier 1–3 bank operators; Tier 4 retail via Local Banks
Loan range	Micro/SME at Local Bank tier; larger exposures via syndication
Key pain points	Opaque reserves; flat DeFi pools; weak explainability in on-chain lending

Table 5.10: Partner categories and roles

Partner Category	Role	Incentive
Financial regulators	Sandbox oversight	On-chain audit trails
Banking institutions	National/Local Bank nodes	Reserve access; lower settlement friction
Oracle providers	Chainlink DON, indexing	Unified oracle stack
Academic partners	ML / formal-verification review	Datasets; co-publication

Market, customer, and partner tables below complement the protocol comparison in Chapter 2 (Table 2.7).

5.7 Discussion

This pre-thesis specifies architecture and planned evaluation; deployment evidence, trained models, and formal proofs arrive after pre-thesis 2. Five examiner-facing limits are recorded explicitly:

- **Novelty is compound, not absolute**—Maple, Goldfinch, and Morpho cover parts of institutional credit; the claim is the four-tier reserve stack plus explainable oracle ML plus group liability in one open design.
- **BCCC is a baseline only**—flat-pool fraud labels may not transfer to hierarchical CWB flows without testnet retraining.
- **Testnet scope**—workshop-grade evidence needs benchmarks and reproducible repos; MVT deployment supplies the empirical layer.
- **Group lending**—on-chain cosigning is not social solidarity; BRAC/Grameen statistics inform design only.
- **Oracle and economics**—scores attach at origination; collusion and model drift remain open; Certora proofs state assumptions explicitly.

The prototype runs on public testnets with test tokens only. Jurisdiction-specific pilots (e.g. regulatory sandboxes) are future work; the architecture is global L2 institutional DeFi, not a single-country rollout claim.

Chapter 6

Conclusion

Authoring instructions — Chapter 6 (short version)

Goal: Close the thesis in 2–4 pages; restate contributions without re-explaining architecture.

Must include:

1. **Opening** (1 paragraph): problem — flat DeFi vs. hierarchical development finance; four-tier proposal in one sentence.
2. **Banking grounding** (1 paragraph): six banking functions encoded on-chain; multi-entity co-lending (solidarity + syndication) as differentiator.
3. **Four contributions** (bullet list): C1 four-tier architecture; C2 group lending; C3 explainable oracle ML; C4 ZKP identity — each one line with honest built-vs-specified scope.
4. **Positioning** (1 paragraph): vs. institutional DeFi literature and MiCA/GENIUS regulatory frontier; no single competitor has full stack.
5. **Pre-thesis status** (1 paragraph): what this report delivers now vs. what MVT adds after pre-thesis 2.

Future work (compress to 1 paragraph or 6 bullets max): hierarchical lending E2E, stablecoin deploy, deposit/group products, ML oracle live, simulation/Certora; optionally one line each on retail rails and GNN/FL research.

Cut from full Ch. 6: repeating Ch. 3–5 detail, long Future Work subsections (retail banking essay, regulatory deep-dive). Keep tone declarative, not promotional.

The *Crypto World Bank* specifies a **four-tier institutional lending architecture** on EVM L2 that flat DeFi protocols (\$55B+ TVL) do not model: reserve-enforced capital flows from World Bank reserve to retail clients, with same-tier interbank pools, upward surplus deposits, and solidarity group lending.

Contributions (scoped): (C1) hierarchical lending + multi-entity contracts; (C2) programmable group liability; (C3) oracle-mediated explainable ML; (C4) privacy-preserving KYC path. The platform encodes six banking functions as auditable on-chain rules rather than discretionary enforcement.

Pre-thesis 1 deliverable: full architecture, literature, methodology, and feasibility fram-

ing; **after pre-thesis 2:** Cardona deploy, E2E loan demo, trained RF/IF on BCCC, populated MVT evidence. Future work: retail payment rails, GNN/federated extensions, Certora proofs, and regulatory pilots.

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